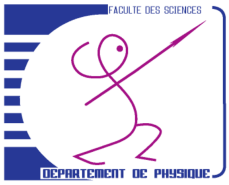


UNIVERSITY OF ANTANANARIVO
FACULTY OF SCIENCES





Field: SCIENCES AND TECHNOLOGIES
Department: PHYSICS AND APPLICATIONS
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A Thesis submitted for the degree of:

MASTER II IN PHYSICS AND APPLICATIONS

Titled:

The Distribution of Kinematically Cold and Warm Atomic Hydrogen (HI) and Its Connection to Star Formation in Nearby Galaxies

Presented on 12 November 2026 by:

Anjia Randia RABARISOA

Before the jury members composed of:

President:	**Mr. Minoson RAKOTOMALALA**	Full Professor
Supervisor:	**Mr. Roger IANJAMASIMANANA**	Posdoctoral Fellow
Examiners:	**Mr. Solohery RANDRIAMAMPANDRY**	Associate Professor
	Mr. Toky RANDRIAMAMPANDRY	Senior Lecturer



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Abstract

The process of star formation within galaxies is fueled by gas from the interstellar medium (ISM), which exists in molecular, atomic and ionised phases. This study investigates the relationship between star formation and the different phases of the ISM in a sample of 16 nearby galaxies, composed of 9 late-type/dwarfs and 7 spirals. We used CO data from *HERACLES*, atomic hydrogen (HI) from *THINGS*, 24 μm from *SINGS*, and UV from *GALEX NGS* surveys. We decomposed the HI emission spectra using the ROHSA algorithm, which takes into account the spatial coherence of neighbouring spectra, to separate kinematically narrow and broad HI components. We use these components as kinematic proxies for the cold neutral medium (CNM) and warm neutral medium (WNM), rather than direct measurements of the thermal phases.

We examine the spatial distribution of the kinematically cold and warm components and find that both components extend across the outermost observed atomic discs. The narrow component contains a median of about 45% of the total HI mass and has a more filamentary appearance than the broad component. Its fractional contribution is not consistently higher in stellar spiral arms, contrary to theoretical expectations that cooling in the arms could favour cold gas formation. In addition, we find no clear evidence that narrow HI is generally more concentrated near CO emissions than broad HI, as would be expected if the narrow component were more closely associated with molecular gas.

We examine the Kennicutt–Schmidt relation for the different gas components to investigate which components are more closely linked to star formation. In dwarf galaxies, the kinematically cold component shows a stronger overall correlation with the SFR than the total HI or the warm component. In spiral galaxies, the warm component is more globally correlated. A possible explanation of this discrepancy is that line width traces different gas conditions in the two types of galaxies. In dwarfs, narrow emission likely traces cooler, less turbulent gas associated with conditions favourable for star formation. In spirals, star-forming regions can contain cold gas whose lines are broadened by turbulence or unresolved motions. Overall, molecular gas remains the component that is more closely linked to the SFR, as previously shown in the literature. The role of kinematically cold atomic gas in the HI-to-H_2 transition remains ambiguous at our sub-kpc resolution.

Keywords: Nearby galaxies, Interstellar medium, Star formation

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List of Abbreviations

BAYGAUD	BAYesian GAUssian Decomposer
BEARS	25-Beam Array Receiver System
CNM	Cold Neutral Medium
CO	Carbon Monoxide
FUV	Far-Ultraviolet
GALEX NGS	Galaxy Evolution Explorer Nearby Galaxies Survey
GMCs	Giant Molecular Clouds
H_2	Molecular Hydrogen
HERACLES	HERA CO-Line Extragalactic Survey
HI	Neutral Atomic Hydrogen
HII	Ionized Hydrogen (H II Region)
HIM	Hot Ionized Medium
IR	Infrared
IRAM	Institut de Radioastronomie Millimétrique
ISM	Interstellar Medium
ISRF	Interstellar Radiation Field
LNM	Lukewarm Neutral Medium
NRAO	National Radio Astronomy Observatory
ROHSA	Regularized Optimization for Hyper-Spectral Analysis
S^4G	Spitzer Survey of Stellar Structure in Galaxies
SFE	Star Formation Efficiency
SFR	Star Formation Rate
SINGS	Spitzer Infrared Nearby Galaxies Survey
SoFiA	Source Finder Application
THINGS	The HI Nearby Galaxy Survey
THOR	The HI/OH/Recombination Line Survey of the Milky Way
VLA	Very Large Array
WIM	Warm Ionized Medium
WNM	Warm Neutral Medium

Nomenclature

Symbol	Nomenclature	Value / Unit
A	Kennicutt–Schmidt law normalization coefficient	-
D	Galaxy distance	Mpc
I	Intensity	MJy sr^{-1}
J	Rotational quantum number	-
λ	Wavelength	m
λ_A, λ_μ, λ_σ, λ''_σ	Hyperparameters for ROHSA	-
N	Kennicutt–Schmidt law power-law index	-
ν	Frequency	Hz
r	Radius	kpc
r_{25}	Optical radius	kpc
ρ	Volume density	g cm^{-3}
σ_v	Velocity dispersion	km s^{-1}
Σ_{gas}	Gas surface density	$M_\odot$ pc^{-2}
Σ_{SFR}	Star formation rate surface density	$M_\odot$ kpc^{-2} yr^{-1}
t_{dep}	Depletion time	yr
X_{CO}	CO-to-H_2 conversion factor	cm^{-2} (K km $s^{-1})^{-1}$
Z	Metallicity	$12+\log(\mathrm{O/H})$ or $0.1\,Z_\odot$

Chapter 1: Introduction

Galaxies are gravitationally bound systems composed primarily of stars, gas, and dust; their evolution is strongly influenced by the interplay between these components. Star formation is a fundamental process in galaxy evolution, regulating the growth of stellar populations and influencing the physical properties of galaxies over cosmic time. The process of star formation within these systems is fuelled by the gas in the interstellar medium (ISM), which fills the vast regions between stars. Stars form within cold, dense molecular clouds composed predominantly of molecular hydrogen (H_2). This molecular phase forms from the diffuse atomic gas through thermodynamic and chemical transitions that remain to be fully understood (Kennicutt, 1998; Klessen & Glover, 2016). Consequently, the rate at which new stars form, commonly quantified as the star formation rate (SFR), is intrinsically linked to the physical properties and thermodynamic state of the interstellar gas across all its phases (Bigiel et al., 2008; Leroy et al., 2008). This thesis examines the link between star formation, the molecular gas and the phases of the HI.

A crucial step in the pathway from diffuse gas to star formation is the transition from atomic to molecular hydrogen. Molecular hydrogen forms primarily through catalytic reactions on the surfaces of interstellar dust grains, as gas-phase formation is inefficient under typical interstellar conditions (Gould & Salpeter, 1963; Hollenbach & Salpeter, 1971). Once formed, H_2 molecules are susceptible to photodissociation by ultraviolet (UV) radiation in the Lyman-Werner bands. The survival of molecular hydrogen therefore largely depends on the shielding from this dissociating radiation, which is provided by dust absorption and by H_2 self-shielding at sufficiently high column densities (Draine & Bertoldi, 1996; Krumholz et al., 2009b; Sternberg et al., 2014).

Theoretical models predict that this shielding becomes effective in regions where sufficient column density has accumulated (Krumholz et al., 2009b; Sternberg et al., 2014). The colder and denser phase of the HI, also known as the cold neutral medium (CNM), with its relatively high density ($\sim$10–100 cm^{-3}), can more readily accumulate the column density required to shield molecules from photodissociation compared to the warmer phase called the warm neutral medium (WNM) ($\sim$ 0.1–1 cm^{-3}) (Krumholz et al., 2009b; Sternberg et al., 2014; Syed et al., 2020). Furthermore, the H_2 formation rate on dust grains scales with the product of atomic hydrogen and dust grain number densities, meaning that denser CNM regions convert atomic gas to molecular form more efficiently

(Hollenbach & McKee, 1979; Wakelam et al., 2017). The CNM thus represents both the environment where shielding conditions are met and the reservoir of atomic gas from which molecular clouds condense (Wolfire et al., 2003; Klessen & Glover, 2016).

Despite the theoretical importance of the CNM as a precursor to molecular cloud formation, empirical studies of the relationship between star formation and the ISM have traditionally focused on either the total HI content or the molecular gas traced by carbon monoxide (CO) emission (Kennicutt, 1998; Bigiel et al., 2008; Leroy et al., 2008). This is largely because separating the CNM from the WNM observationally has been challenging. The two phases contribute to the same 21 cm emission line and are not straightforward to distinguish observationally, with absorption measurements against background continuum sources preferentially tracing the cold component but providing only a sparse spatial sampling, while spectral decomposition techniques offer an alternative approach that has only recently become sophisticated enough for systematic application to external galaxies (Heiles & Troland, 2003a; Marchal et al., 2019; Syed et al., 2020).

Understanding the relationships between the different ISM phases and star formation is essential for describing the evolution of galaxies. Nearby galaxies serve as ideal laboratories for such investigations, as they can be observed in their entirety, unlike our own Milky Way where our position within the disc limits our view (Leroy et al., 2008). Moreover, establishing quantitative relationships between the CNM and molecular gas would provide a valuable diagnostic tool for studying star formation in environments where CO observations are unavailable or unreliable. This is particularly relevant for low-metallicity dwarf galaxies, where reduced dust shielding leads to more efficient CO photodissociation, yet H_2 may still persist in regions where self-shielding is effective (Bolatto et al., 2013).

While power-law relations between gas surface density and star formation rate surface density, commonly known as the Kennicutt-Schmidt laws, remain one of the most widely used empirical frameworks for relating gas and star formation (Schmidt, 1959; Kennicutt, 1998), the role of individual ISM phases in these relations is still poorly constrained by observations. In their most common form, these relations treat the neutral gas as a single reservoir, summing HI and H_2 into one total gas surface density, which obscures which gas phase is actually driving them. Resolved THINGS observations show that the star formation rate surface density scales roughly linearly with Σ_{H_2} alone, while HI saturates near 9–10 $M_\odot\,pc^{-2}$ and correlates only weakly with star formation once that threshold is reached (Bigiel et al., 2008). Simulations of the cold neutral medium in spiral discs find, in turn, that the star formation rate density follows a scaling law with CNM column density similar to the total-gas Kennicutt-Schmidt law, with the CNM tracing the star-forming gas more closely than the neutral atomic gas as a whole (Smith et al., 2023). The goal of this study is to provide a comprehensive analysis

of the relationship between star formation rate surface density and gas surface density in a sample 16 of nearby galaxies, with particular emphasis on examining whether the kinematically cold component of HI can serve as an effective tracer for the conditions that precede molecular cloud formation and star formation. By applying modern spectral decomposition techniques to high-resolution HI data cubes, this work aims to quantify how kinematically cold HI, which serves as a proxy for the thermally defined CNM, correlates with molecular gas content and star formation activity. Ultimately, this study seeks to test the theoretical prediction that CNM abundance governs the efficiency of the HI-to-H_2 transition and subsequent star formation.

1.1 The interstellar medium

1.1.1 Overview

The space between stars in galaxies is not devoid of matter but is filled with a complex mixture of gas, dust, relativistic charged particles (cosmic rays), and electromagnetic fields. This diffuse material constitutes the interstellar medium (Ward-Thompson & Whitworth, 2011; Ferrière, 2001).

The gaseous component dominates the mass budget of the ISM and is typically concentrated in cloud-like structures of varying sizes and morphologies (Ward-Thompson & Whitworth, 2011). Figure 1a, for example, showcases a molecular hydrogen (H_2) cloud. Hydrogen is the most abundant element in galaxies and exists in several forms: atomic (commonly denoted HI), molecular (H_2), and ionised (HII). In the Milky Way, the ISM is composed of approximately 70% hydrogen, 28% helium, and 2% heavier elements by mass (Ferrière, 2001). In astronomical nomenclature, all elements heavier than helium are collectively referred to as "metals", including carbon, nitrogen, and oxygen.

Interstellar dust consists of small solid grains, typically less than a micrometre in size, predominantly composed of silicates, carbonaceous materials, and iron-bearing compounds (Zhukovska et al., 2008). Some molecular gas clouds appear as dark patches against the stellar background because their dust content is opaque to visible light, absorbing and scattering optical radiation from background stars. The Coalsack Nebula (Figure 1b) provides a striking example of such "dark clouds" or "absorption nebulae" (Ward-Thompson & Whitworth, 2011).

There is significant scientific interest in understanding the properties and dynamics of the ISM because the life cycles of stars and interstellar matter are intimately connected. Stars form from the gravitational collapse of interstellar gas clouds. Throughout their main-sequence lifetime, stars

fuse hydrogen into helium in their cores. At the end of stellar evolution, when nearly all hydrogen fuel is exhausted, stars either expand and eject their outer layers as planetary nebulae or, for more massive progenitors, explode as supernovae. In either scenario, processed material enriched with heavy elements is returned to the ISM, becoming available as raw material for subsequent generations of stars (Ward-Thompson & Whitworth, 2011). The dynamics of the ISM therefore determine where and when stars are born within a galaxy, and the properties of newly formed stars and their planetary systems are closely linked to the chemical and physical conditions of their natal environments.

1.1.2 Phases of the gaseous interstellar medium

The gaseous ISM is far from homogeneous; it exhibits vastly different physical properties depending on the location within a galaxy. Due to this complexity, no single classification scheme adequately captures all its characteristics. Various studies have proposed different frameworks for describing the ISM phases. Klessen & Glover (2016) described five distinct phases, illustrated in the table below:

Table 1: ***Description and typical physical properties of the ISM.*** *Values compiled from Klessen & Glover (2016) and Ferrière (2001).*

Phase	Description	Temperature [K]	Density [cm^{-3}]	Ionisation
Dense cold molecular gas (H_2)	Cold, dense gas consisting primarily of molecular hydrogen (H_2).	10–20	$> 10^2$	$< 10^{-6}$
Cold Neutral Medium (CNM)	Cold, dense, primarily neutral atomic gas.	50–100	20–50	10^{-4}
Warm Neutral Medium (WNM)	Warm, diffuse, neutral atomic gas.	6000–10000	0.2–0.5	≈ 0.1
Warm Ionised Medium (WIM)	Warm, diffuse, (partially to fully) ionised gas.	~ 8000	0.2–0.5	1.0
Hot Ionised Medium (HIM)	Hot, diffuse, ionised gas.	$\sim 10^6$	$\sim 10^{-2}$	1.0

Cold and warm neutral medium

When neutral atomic hydrogen (HI) is in thermal equilibrium, meaning its temperature remains stable due to a balance between heating and cooling processes, it exists in two thermally stable phases: the CNM and the WNM (Field et al., 1969; Wolfire et al., 2003). Dense clouds of CNM are commonly found embedded within a diffuse WNM envelope.

Gas with temperatures intermediate between these two stable phases is termed the lukewarm

neutral medium (LNM) or thermally unstable neutral medium (UNM). This gas is thermally unstable and will eventually either increase in density and cool to become part of the CNM, or become more diffuse and heat up to join the WNM (Field et al., 1969; Bialy & Sternberg, 2019). The existence of significant amounts of gas in this thermally unstable regime has been confirmed observationally (Heiles & Troland, 2003b; Kanekar et al., 2003; Murray et al., 2018).

The WNM at its typical density cannot spontaneously transition into CNM. However, an isothermal compression of the WNM by a factor of two to four is sufficient to induce a phase transition. In other words, CNM forms when WNM experiences increased pressure and enters a thermally unstable phase. Such transitions can be driven by transient phenomena including supernova explosions, stellar winds, or large-scale turbulent compressions (Saury, E. et al., 2014; Klessen & Glover, 2016).



(a) ***NGC 2174, or Monkey Head Nebula*** *(b)* ***Coalsack Nebula***

Figure 1: Left: NGC 2174, also known as the Monkey Head Nebula, showcasing the cloud-like formation of H_2 gas. Image credit: Hubble Space Telescope. Right: The Coalsack Nebula, a molecular cloud in the constellation Crux, appearing as a dark patch when observed with the naked eye. Image credit: ESO.

Molecular gas

The molecular phase of the ISM is dominated by H_2, though it also contains simple molecules such as carbon monoxide (CO) and water (H_2O), as well as more complex organic species including methanol (CH_3OH) and sugar-related molecules like glycolaldehyde (CH_2OHCHO) (Hollis et al., 2000). Molecular clouds are the densest regions of the ISM, with typical number densities exceeding 10^2 cm^{-3} (Table 1) and can contain dust grains. Despite being the densest ISM phase, this density remains far lower than terrestrial atmospheric conditions, which are on the order of 10^{19} particles

per cubic centimetre. The largest molecular gas structures are referred to as Giant Molecular Clouds (GMCs), and in galaxies, these GMCs serve as the principal sites of star formation (Klessen & Glover, 2016).

Warm and hot ionised medium

The warm ionised medium (WIM) and hot ionised medium (HIM) represent the ionised phases of the ISM. The WIM has temperatures of approximately 8000 K and is maintained by ionising radiation from massive stars, while the HIM reaches temperatures of $\sim 10^6$ K and is primarily heated by supernova explosions (Ferrière, 2001). Although these phases are important components of the ISM ecology, they are not the primary focus of this study, which concentrates on the neutral atomic and molecular phases most directly relevant to star formation.

1.1.3 Tracing the phases of neutral atomic hydrogen

The properties of the ISM can be studied through observations of emission and absorption lines from different atomic and molecular species. Emission lines arise when electrons in atoms transition from higher to lower energy states, releasing photons at characteristic frequencies determined by the energy difference between levels. Absorption lines occur when atoms absorb photons of specific frequencies, causing electrons to transition to higher energy states. Since energy levels are quantised, each element and molecule possesses unique spectral signatures.

The 21 cm hyperfine transition

Neutral atomic hydrogen (HI) can be observed through its 21 cm wavelength hyperfine transition, which lies in the radio portion of the electromagnetic spectrum. The electronic ground state of the hydrogen atom is split into two closely spaced energy levels (Figure 2): a higher energy state where the spins of the electron and proton are parallel (excited state, $S = 1$), and a lower energy state where they are anti-parallel (ground state, $S = 0$). A transition between these levels, commonly known as the "spin-flip" transition, corresponds to an energy difference of $\Delta E = 5.87 \times 10^{-6}$ eV, producing a photon with frequency $\nu = 1.4204$ GHz or wavelength $\lambda = 21.106$ cm (Ward-Thompson & Whitworth, 2011). Unlike CO(1–0), which is optically thick and confined to the dense, star-forming molecular phase, 21 cm emission is essentially optically thin across most of the interstellar medium and traces the diffuse atomic gas far from sites of active star formation, allowing it to be detected well beyond the

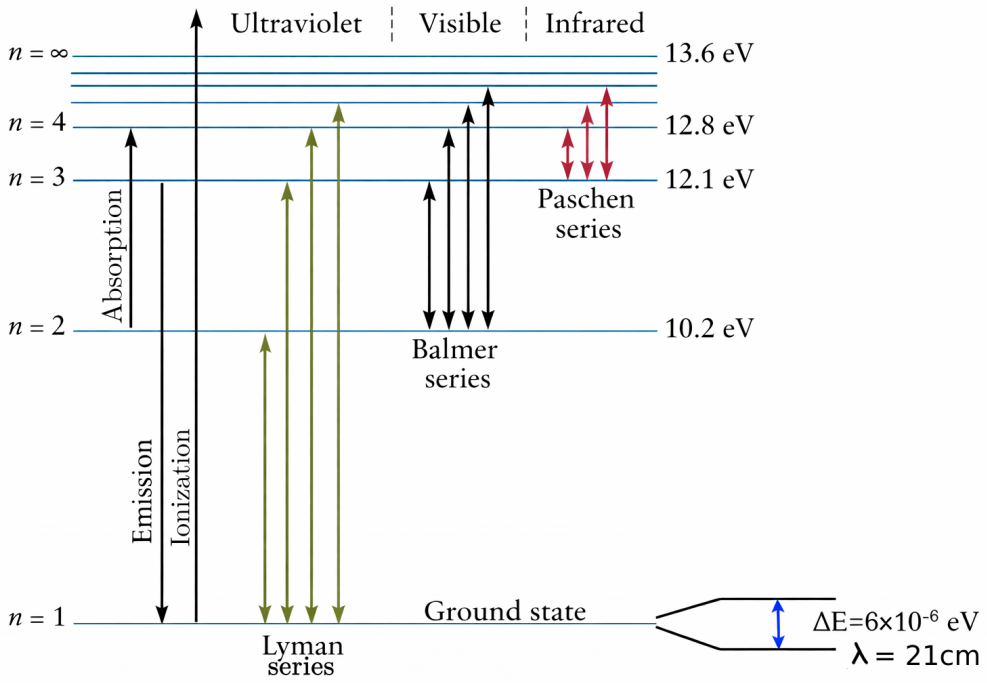


Figure 2: The energy level diagram of the hydrogen atom. n represents the principal quantum number of the electron orbital. ΔE indicates the energy of the spin-flip transition responsible for the 21 cm emission line. Image credit: DARA Foundation Astrophysics - Interstellar Gas part 2.

stellar disc where the gas is too diffuse or too weakly shielded to host a molecular phase. Deep 21 cm surveys have repeatedly confirmed that HI discs extend well beyond the optical disc: the HI radius at which the surface density falls to 1 $M_\odot\,pc^{-2}$, R_{HI}, exceeds the optical radius R_{25} by a factor of ~ 1.5 on average, with a tail towards considerably larger ratios (Broeils & Rhee, 1997; Wang et al., 2016), and recent WALLABY pilot observations of a statistical sample of local galaxies find that, on average, only about two-thirds (68%) of the total HI mass is contained within R_{25}, with the remainder residing in low surface density gas at larger radii (Lee et al., 2025). This extended atomic gas reservoir, largely inert to star formation but dynamically coupled to the disc, has proven a key tracer of galaxy interactions, accretion, and the outer mass distribution that optical observations alone cannot access.

Absorption-emission techniques for phase separation

The classical method for determining the physical properties of HI, particularly distinguishing between the CNM and WNM, combines 21 cm emission and absorption observations (Kulkarni & Heiles, 1987; Dickey & Lockman, 1990; Heiles & Troland, 2003a). This technique relies on the fact that absorption is primarily sensitive to the CNM due to its high optical depth, which results from the combination of high density and low temperature relative to the WNM (Kanekar et al., 2003; Roy et al., 2013).

In the two-phase model of the ISM, the spin temperature T_s (which characterises the relative populations of the two hyperfine levels) is related to the kinetic temperature T_k of the gas. For the CNM, collisions are sufficiently frequent to thermalise the hyperfine levels, resulting in $T_s \approx T_k$ in the range 40–200 K. For the WNM, the lower density means that collisions alone cannot thermalise the line, and T_s may be significantly lower than T_k unless Lyman α radiation couples the spin and kinetic temperatures (Field, 1958; Liszt, 2001; Heiles & Troland, 2003b).

Heiles & Troland (2003a,b) conducted an extensive absorption-emission survey (the Millennium Arecibo 21 Centimeter Absorption-Line Survey) of 79 continuum sources and developed sophisticated Gaussian decomposition methods to separate CNM and WNM components. They found that the CNM spin temperature histogram peaks at approximately 40 K, and that about 60% of all HI resides in the WNM phase. Importantly, they also found evidence that a significant fraction of the WNM (at least 48%) lies in the thermally unstable temperature range of 500–5000 K.

However, observing absorption lines requires sufficiently bright, compact radio continuum sources in the background, and the scarcity of such sources severely limits the spatial coverage achievable with this method, particularly for extragalactic observations (Kanekar et al., 2003; Murray et al., 2018). For the Milky Way, large absorption surveys such as 21-SPONGE (Murray et al., 2018) and THOR (Beuther et al., 2016) have expanded the number of sightlines probed, but mapping the CNM and WNM phases across entire external galaxies remains impractical using absorption techniques alone.

Gaussian decomposition of emission spectra

Given the limitations of absorption observations, alternative approaches have been developed to extract phase information from HI 21 cm emission spectra alone. These methods typically rely on the assumption that the CNM and WNM have characteristically different velocity dispersions: CNM components are kinematically narrow ($\sigma_v \sim$ 1–6 km s^{-1}) due to their low temperatures, while WNM components are kinematically broad ($\sigma_v \sim$ 6–15 km s^{-1}) due to their higher thermal velocities and turbulent motions (Young et al., 2003; de Blok & Walter, 2006; Ianjamasimanana et al., 2012).

Early studies demonstrated that these kinematically distinct components can be identified from HI emission profiles. Young et al. (2003) fitted Gaussian components to the HI velocity profiles of seven nearby dwarf galaxies and identified narrow components with $\sigma_v \sim$ 3–5 km s^{-1} and broad components with $\sigma_v \sim$ 8–13 km s^{-1}. The narrow components showed a clumpy distribution and tended to be located near star formation regions, whereas the broad components showed a more ubiquitous

distribution. de Blok & Walter (2006) subsequently fitted two-component Gaussian models to HI profiles in NGC 6822, selecting preferred models based on reduced χ^2 values. They likewise identified narrow and broad components, with mean velocity dispersions of ~ 4 and ~ 8 km s^{-1}, respectively.

However, fitting individual HI profiles can be problematic because typical line-of-sight velocity profiles have low to moderate signal-to-noise ratios, making their shapes susceptible to noise. To overcome this limitation, Ianjamasimanana et al. (2012) introduced the "super-profile" technique, in which individual HI velocity profiles from The HI Nearby Galaxy Survey (THINGS) are aligned in velocity and stacked to construct high signal-to-noise composite profiles. By fitting these super-profiles with double Gaussian models, they identified narrow ($\sigma_v \sim 3.4$–8.6 km s^{-1}, average 6.5 ± 1.5 km s^{-1}) and broad ($\sigma_v \sim 10.1$–24.3 km s^{-1}, average 16.8 ± 4.3 km s^{-1}) components, which they interpreted as evidence for the presence of CNM and WNM phases.

Subsequent work extended the super-profile analysis to examine variations in the properties of the narrow and broad components. Ianjamasimanana et al. (2015) stacked individual HI velocity profiles and decomposed them into broad and narrow Gaussian components, studying the HI velocity dispersion and surface density as a function of radius in 22 galaxies from THINGS.

A further development focused on making Gaussian decomposition of individual profiles more robust and fully automated. Oh et al. (2019) introduced BAYGAUD (BAYesian GAUssian Decomposer), a Bayesian Markov Chain Monte Carlo algorithm that fits multiple Gaussian models to individual line-of-sight velocity profiles and selects the optimal number of components using Bayesian model selection. The decomposed components can then be classified as kinematically cold or warm based on their velocity dispersions. Unlike conventional χ^2-based fitting, BAYGAUD reduces the sensitivity to initial parameter estimates and allows profile decomposition and component classification to be performed in a fully automated manner, making the approach suitable for large HI surveys.

Park et al. (2022) applied BAYGAUD to NGC 6822 and found that kinematically cold HI components correlate with regions of active star formation. Building on this Bayesian decomposition approach, Kim & Oh (2022) introduced a new method for constructing HI super-profiles in which the velocity profiles are first decomposed into an optimal number of Gaussian components using BAYGAUD, after which the decomposed components are aligned in velocity and co-added. This approach was designed to reduce the effects of asymmetric velocity profiles, turbulent gas motions, and

beam smearing on conventional super-profiles, while retaining the improved signal-to-noise provided by stacking.

More recently, Park et al. (2025) explored the potential of kinematically colder HI components (defined as those with $\sigma_v < 6$ km s^{-1}) as tracers for star-forming gas in seven nearby galaxies spanning a range of metallicities. They found that these narrow components are often clumpier than broader components and tend to be morphologically associated with regions of active star formation, though the correlation may break down at high physical resolutions where spatial offsets between gas and star formation become apparent.

Kinematically cold versus thermally cold hydrogen

It is important to distinguish between "kinematically cold" and "thermally cold" HI. The thermally defined CNM and WNM are characterised by their physical temperatures, which can only be directly measured through absorption-emission techniques that constrain the spin temperature. In contrast, kinematic decomposition methods identify components based on their velocity dispersions, which reflect a combination of thermal broadening, turbulent motions, and non-thermal processes.

While narrow velocity dispersions are generally associated with cold gas (since thermal broadening scales with temperature), several factors can complicate this interpretation. Intrinsically narrow lines can be broadened by turbulent motions, and the same apparent effect can arise artificially from beam smearing in galaxies with ordered rotation (Ianjamasimanana et al., 2012; Caldú-Primo et al., 2013; Stilp et al., 2013). Furthermore, multiple cold clouds at different velocities along the same line of sight can produce composite profiles that appear broader than individual components.

Throughout this study, the terms "kinematically cold" and "kinematically warm" HI are preferred when referring to components identified through Gaussian decomposition of emission spectra, to acknowledge that these may not correspond exactly to the thermally defined CNM and WNM phases. This distinction is particularly important when interpreting correlations with star formation, as the physical connection between narrow-line HI and the conditions necessary for molecular cloud formation involves complex processes including thermal instability, dust shielding, and gravitational collapse (Schaye, 2004; Krumholz et al., 2011; Klessen & Glover, 2016).

Tracing molecular hydrogen

Unlike HI, direct observation of H_2 at the typical temperatures found in molecular clouds ($\sim$10–40 K) is extremely challenging. As a homonuclear diatomic molecule, H_2 lacks a permanent electric dipole moment and has a low moment of inertia. Its lowest rotational transitions require excitation temperatures of approximately 540 K, far exceeding typical molecular cloud temperatures (Ferrière, 2001; Ward-Thompson & Whitworth, 2011).

One approach to directly detect H_2 is through its ultraviolet absorption lines, but this requires a sufficiently bright hot star aligned along the line of sight, making large-scale mapping impractical. Instead, carbon monoxide (CO) is commonly used as a tracer for molecular gas. CO is relatively abundant, and its rotational transitions can be easily excited at low temperatures. The molecular hydrogen column density is derived from CO observations using a conversion factor, commonly denoted $X_{\rm CO}$ or $\alpha_{\rm CO}$, which depends on environmental factors including metallicity (Bolatto et al., 2013).

This method has important limitations, particularly in low-metallicity environments. Since CO is more easily photodissociated than H_2 by ultraviolet radiation, it survives only in the densest, most shielded regions of molecular clouds. This leaves a significant fraction of H_2 untraced by CO observations, creating what is termed "CO-dark" molecular gas (Bolatto et al., 2013; Wolfire et al., 2010). This effect is most pronounced in the outer regions of molecular clouds and in low-metallicity galaxies where dust shielding is less effective.

1.2 Star formation and the interstellar medium

1.2.1 Overview of the star formation process

Stars form within molecular clouds and GMCs when gravitational collapse overcomes the internal pressure support of the gas. The condition for self-gravitating collapse can be characterised by the Jeans length (r_J), which represents the minimum radius at which a spherical gas cloud of uniform density becomes gravitationally unstable:

$$r_J \sim \frac{a_0}{(G\rho_0)^{1/2}} \tag{1.1}$$

where G is the gravitational constant, ρ_0 is the initial gas density, and a_0 is the sound speed. Since the Jeans length is inversely proportional to the square root of density, gravitational collapse can be

triggered by external compression events such as nearby supernova explosions, shock waves, cloud-cloud collisions, or galaxy interactions. Alternatively, collapse may proceed spontaneously when a cloud's mass exceeds the Jeans mass (Jeans, 1902; Ward-Thompson & Whitworth, 2011).

During collapse, portions of the contracting cloud may themselves become gravitationally unstable, leading to fragmentation into smaller, denser substructures. Each fragment continues to contract until it becomes sufficiently opaque to trap the energy released by cooling processes. This leads to the formation of a hot, pressure-supported core that continues to contract, eventually forming a protostar (Ward-Thompson & Whitworth, 2011; Klessen & Glover, 2016).

1.2.2 The HI-to-H_2 transition and its role in star formation

Since the diffuse neutral ISM is predominantly composed of HI while star-forming GMCs consist mainly of H_2, there must be a transition stage in cloud evolution where atomic hydrogen converts to molecular hydrogen. This HI-to-H_2 transition plays a fundamental role in the evolution of the ISM and the process of star formation (Lee et al., 2012; Sternberg et al., 2014; Klessen & Glover, 2016).

The star formation rate correlates strongly with the HI-to-H_2 transition (Krumholz et al., 2011), because the conditions required for H_2 formation are similar to those needed for star formation. For gas to collapse and form stars, it must cool sufficiently to overcome thermal pressure support. However, the interstellar radiation field (ISRF) heats the gas, counteracting this cooling, and ultraviolet photons also dissociate H_2 molecules. H_2 forms mainly on the surfaces of dust grains, and its formation is faster at higher gas densities (Glover & Mac Low, 2007). Dust grains therefore play a dual role: they catalyse the HI-to-H_2 transition and shield the gas from the ISRF, preventing heating while protecting H_2 from photodissociation (Klessen & Glover, 2016). Molecular gas accumulates only where dust and H_2 itself absorb enough radiation to protect the molecules, so the transition depends on the gas density, the shielding column and the dust abundance (Krumholz et al., 2009a; Sternberg et al., 2014). While metal-line cooling dominates in metal-rich regions, H_2 can also serve as a coolant in low-metallicity environments ($< 0.1 Z_{\odot}$) or where the ISRF is weak, enabling star formation even in primordial or metal-poor clouds (Glover & Clark, 2014).

1.2.3 Empirical star formation laws

As discussed in the previous sections, the ISM provides the fuel for star formation, with its different phases contributing in various ways to the process. Understanding the relationship between the ISM

and star formation is crucial for developing accurate models of galaxy evolution. However, on galactic scales, the complexity of the relevant physical processes precludes purely theoretical predictions. Since star formation correlates well with gas properties, it is common to describe the link between star formation and the ISM using empirical power-law relations, often referred to as star formation laws.

Schmidt (1959, 1963) proposed a power-law relation between the volume star formation rate density f and the gas volume density $\rho_{\rm gas}$:

$$f \propto \rho_{\rm gas}^{n} \tag{1.2}$$

Subsequently, one of the most widely used formulations relates the star formation rate per unit area (the SFR surface density, $\Sigma_{\rm SFR}$) to the gas surface density ($\Sigma_{\rm gas}$):

$$\Sigma_{\rm SFR} = A\,(\Sigma_{\rm gas})^{N} \tag{1.3}$$

This relation is commonly referred to as the Kennicutt-Schmidt law. Here, N is the power-law index (or slope), and A is the normalisation coefficient. Surface densities have the practical advantage of being directly observable for external galaxies, unlike volume densities which require assumptions about the gas distribution along the line of sight. $\Sigma_{\rm SFR}$ is typically measured in $M_{\odot}\,{\rm kpc}^{-2}\,{\rm yr}^{-1}$ and $\Sigma_{\rm gas}$ in $M_{\odot}\,{\rm pc}^{-2}$.

The value of N has physical implications for how star formation is regulated. A value of $N = 1$ implies that the SFR increases linearly with gas availability. For $N > 1$, higher density regions form stars more efficiently than lower density regions, while $N < 1$ indicates diminishing star formation efficiency at higher gas densities.

Kennicutt (1998) studied the correlation between disc-averaged $\Sigma_{\rm SFR}$ and $\Sigma_{\rm HI+H_2}$ for a sample of 61 nearby normal spiral galaxies and 36 starburst galaxies. He found $N = 2.47 \pm 0.39$ for normal spirals alone and $N = 1.4 \pm 0.15$ for the combined sample, later updated to $N = 1.41 \pm 0.07$ by de los Reyes & Kennicutt (2019). The value $N \approx 1.4$ has been widely adopted as a baseline for many star formation, galaxy evolution, and cosmological simulations (Schaye & Dalla Vecchia, 2008; Yajima et al., 2022; Schaye et al., 2023).

The power-law index varies depending on the spatial scale, galaxy type, and gas component considered. Boissier et al. (2003) found $N \approx 2$ for 16 galaxies. For the Milky Way, Misiriotis et al. (2006) derived $N = 2.18 \pm 0.20$, while for M33, Heyer et al. (2004) obtained $N = 3.3 \pm 0.1$. When considering molecular gas alone, Leroy et al. (2005) found $N = 1.3 \pm 0.1$ for CO in dwarf galaxies, and Bigiel et al. (2008) showed that for spiral galaxies $N = 1.0 \pm 0.2$ for $\Sigma_{\rm H_2}$, suggesting a nearly

linear relationship between molecular gas and star formation.

An alternative formulation of the ISM-SFR relationship is the star formation efficiency (SFE):

$$\mathrm{SFE} = \frac{\Sigma_{\mathrm{SFR}}}{\Sigma_{\mathrm{gas}}} \tag{1.4}$$

The SFE is measured in yr^{-1} and can be interpreted as the fraction of gas converted into stars per unit time. It is also the inverse of the gas depletion time ($t_{\mathrm{dep}} = 1/\mathrm{SFE}$), which represents the time required for ongoing star formation to consume the entire gas reservoir (Leroy et al., 2008; Teng et al., 2024).

1.2.4 Star formation thresholds and the role of cold gas

The presence of gas alone does not guarantee star formation. Observationally, star formation follows a monotonic trend at high gas densities but exhibits a sharp decline at low densities. There exists a critical surface density (Σ_{crit}), often referred to as the star formation threshold, below which gas cannot efficiently form stars. This threshold has been observed to lie in the range 5–10 $M_{\odot}\,\mathrm{pc}^{-2}$ (Kennicutt, 1998; Martin & Kennicutt, 2001).

Various physical mechanisms have been proposed to explain these thresholds, including large-scale gravitational instabilities, shear, metallicity effects, and turbulence (Leroy et al., 2008; Sternberg et al., 2014; Federrath, 2015). Of particular relevance to this study, Schaye (2004) demonstrated that the drop in thermal velocity dispersion associated with the transition from WNM to CNM triggers gravitational instability, leading to cloud collapse and star formation. He proposed that the environmental conditions where HI is typically cold, which leads to thermogravitational instability, can serve as a star formation threshold, predicting $\Sigma_{\mathrm{crit}} \approx 3$–$10\ M_{\odot}\,\mathrm{pc}^{-2}$.

Since previous studies relied on the assumption that CNM is found where H_2 formation is efficient, this work aims to test that hypothesis directly by examining correlations between star formation and the individual gas phases, with particular emphasis on the kinematically cold component of the HI. To better understand the relationship between ISM properties and star formation, the Kennicutt-Schmidt relation will be applied to different gas components. Additionally, SFE for each gas component will be studied.

Chapter 2: Methodology and data

This chapter describes the observational data, the methods used to separate and map the HI components, the fitting methods, and the galaxy sample.

2.1 Neutral atomic hydrogen

All HI data used in this study were obtained from *The HI Nearby Galaxy Survey* (THINGS; Walter et al. 2008), which used the Karl G. Jansky Very Large Array (VLA) telescope of the National Radio Astronomy Observatory (NRAO)[1].

The VLA is located on the Plains of San Agustin in New Mexico and consists of 27 antennas, each with a diameter of 25 metres, arranged in a Y-shaped configuration with a maximum baseline of 35 km. The array covers frequencies ranging from 50 MHz to 97 GHz, achieving angular resolutions as fine as 0.1 arcseconds and sensitivities of approximately 100 microjanskys (Thompson et al., 1980).

THINGS surveyed the 21 cm emission line in 34 galaxies in the local neighbourhood of the Milky Way, with the goal of investigating the fundamental nature and characteristics of the ISM in relation to galaxy morphology, star formation, and mass distribution across the Hubble sequence. All 34 galaxies are situated at distances between 2 and 15 Mpc. THINGS mapped HI emission in 34 nearby galaxies, with the goal of investigating the properties of the ISM in relation to galaxy morphology, star formation, and mass distribution. All 34 galaxies are located at distances between 2 and 15 Mpc. The survey provides two sets of data cubes with different weighting schemes: robust weighting, optimized for spatial resolution ($\sim 6''$), and natural weighting, optimized for sensitivity to diffuse emission ($\sim 11''$). The velocity resolution is 1.3 km s^{-1} for two galaxies and either 2.6 or 5.2 km s^{-1} for the rest (Walter et al., 2008). The sample spans a wide range of properties: SFR from 10^{-3} to 6 $M_\odot$ yr^{-1}, total HI mass from 0.01 to $14 \times 10^9 M_\odot$, absolute B-band magnitude from -11.5 to -21.7 mag, and metallicities from 7.5 to 9.2 in units of $12 + \log[\mathrm{O/H}]$. The THINGS data products include spectral cubes, integrated HI intensity maps (moment 0), velocity maps (moment 1), and velocity dispersion

[1]The National Radio Astronomy Observatory is a facility of the National Science Foundation operated under cooperative agreement by Associated Universities, Inc.

maps (moment 2), all publicly available online[2].

The HI surface density $\Sigma_{\rm HI}$ was derived from the moment 0 maps using the relation from Leroy et al. (2008):

$$\Sigma_{\rm HI} = 0.020 \times I_{21\rm cm} \times \cos i \tag{2.1}$$

where $\Sigma_{\rm HI}$ is in $M_\odot$ pc^{-2}, $I_{21\rm cm}$ is the integrated intensity in K km s^{-1}, and i is the inclination angle of the galaxy. The factor $\cos i$ corrects for the projection of the disc onto the plane of the sky. The left column of Figure 3 shows a few example $\Sigma_{\rm HI}$ maps from the galaxies in this study.

2.2 Star formation rate

The SFR surface density maps were constructed using a combination of mid-infrared (24 μm) and far-ultraviolet (FUV) data. The 24 μm observations were obtained from the *Spitzer Infrared Nearby Galaxies Survey* (SINGS; Kennicutt et al. 2003) using the *Spitzer Space Telescope*, while the FUV data came from the *Galaxy Evolution Explorer Nearby Galaxies Survey* (GALEX NGS; Gil de Paz et al. 2007). The SINGS data have an angular resolution of 6 arcseconds, and the GALEX NGS data have a resolution of 4.0 arcseconds.

The 24 μm emission primarily originates from dust grains heated by ultraviolet radiation produced by young, massive stars (predominantly O and B types) in star-forming regions. These hot stars emit copious UV photons, which are absorbed by surrounding dust grains, heating them. The dust then re-emits this absorbed energy in the infrared (Calzetti et al., 2007).

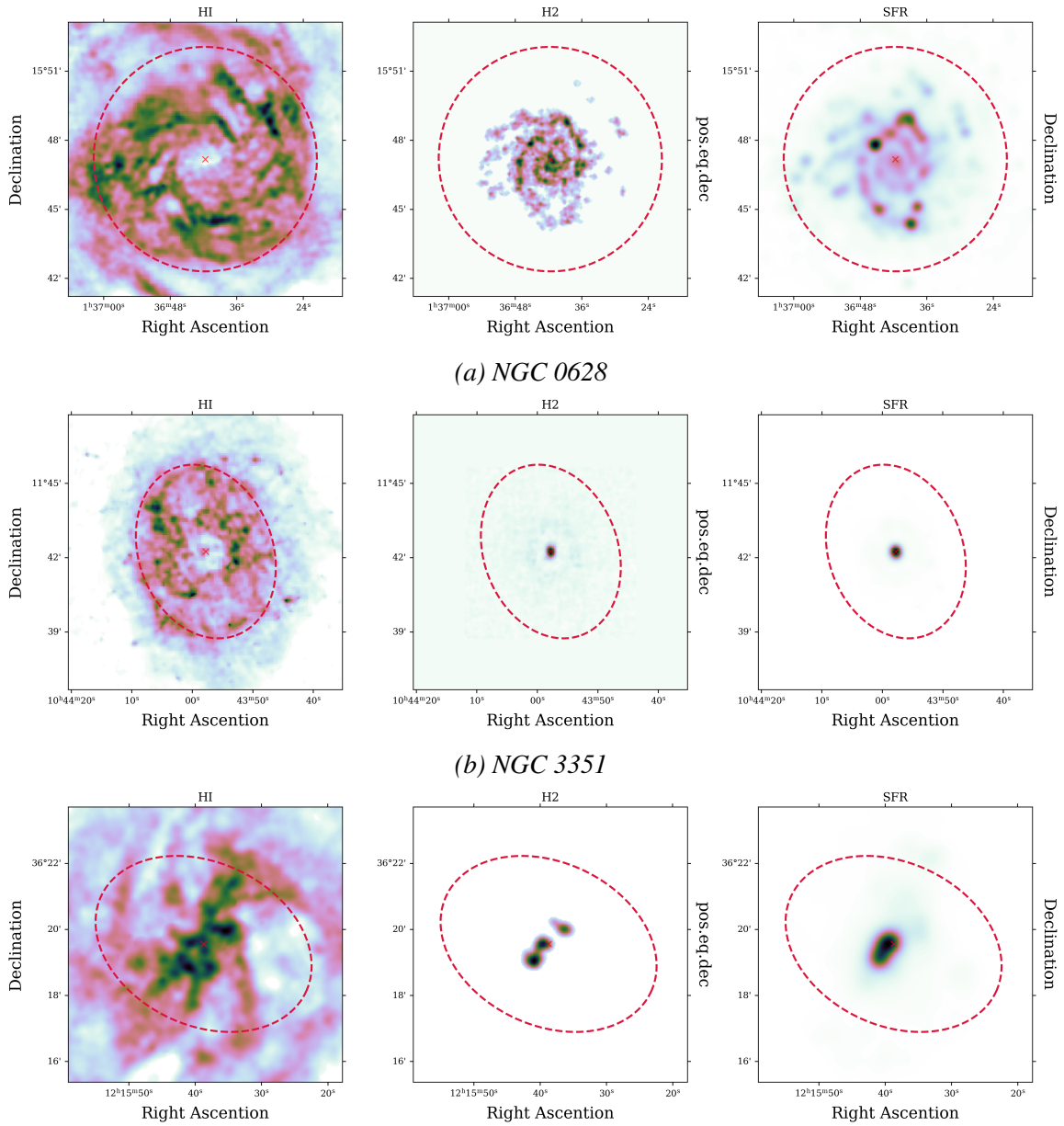
FUV emission, by contrast, traces unobscured star formation, as the light comes directly from the photospheres of young massive stars. Since these stars dominate the FUV luminosity, the FUV flux serves as a sensitive indicator of recent star formation, typically averaged over the past $\sim$100 Myr (Salim et al., 2007).

Combining these two tracers provides a more complete census of star formation: the FUV captures unobscured star formation while the 24 μm emission accounts for dust-obscured activity. The SFR surface density $\Sigma_{\rm SFR}$ was calculated using the calibration from Leroy et al. (2008), which was derived from the 24 μm and FUV-to-SFR calibrations of Calzetti et al. (2007) and Salim et al. (2007):

$$\Sigma_{\rm SFR} = (8.1 \times 10^{-2} I_{\rm FUV} + 3.2^{+1.2}_{-0.7} \times 10^{-3} I_{24\mu\rm m}) \cos i \tag{2.2}$$

[2]http://www.mpia.de/THINGS

where $\Sigma_{\rm SFR}$ is in $M_\odot$ yr^{-1} kpc^{-2}, $I_{\rm FUV}$ and $I_{24\mu{\rm m}}$ are the FUV and 24 μm intensities in MJy sr^{-1}, and i is the galaxy inclination angle. A sample of $\Sigma_{\rm SFR}$ maps is shown in the right column of Figure 3.



(a) NGC 0628

(b) NGC 3351

(c) NGC 4214

Figure 3: Example maps of the surface densities of HI (left column), H_2 (middle column) and SFR (right column) from the galaxies in this study. The red dashed annuli display the optical radius r_{25}. The cross indicates the centre of each galaxy.

2.3 Molecular hydrogen

We use carbon monoxide (CO) maps from the *Heterodyne Receiver Array CO Line Extragalactic Survey* (HERACLES; Leroy et al. 2009) to trace molecular hydrogen (H_2). HERACLES used the

30-metre telescope of the *Institut de Radioastronomie Millimétrique* (IRAM)[3], located on Pico Veleta in the Spanish Sierra Nevada. The survey mapped the CO J=2–1 emission line at 230 GHz in 48 nearby galaxies. The HERACLES maps have excellent spatial coverage, extending to the optical radius r_{25} for each target, with an angular resolution of 13 arcseconds and a velocity resolution of 2.6 km s^{-1}. The data products include CO datacubes, integrated CO intensity maps (moment 0), and intensity-weighted velocity maps (moment 1), which are publicly available online[4].

The H_2 surface density Σ_{H_2} was derived from the integrated CO intensity I_{CO} using the relation from Leroy et al. (2008) for the J=2–1 transition:

$$\Sigma_{H_2} = 5.5 \times I_{CO} \times \cos i \tag{2.3}$$

or for the J=1–0 transition:

$$\Sigma_{H_2} = 4.4 \times I_{CO} \times \cos i \tag{2.4}$$

where Σ_{H_2} is in $M_\odot$ pc^{-2}, I_{CO} is in K km s^{-1}, and i is the inclination angle. These relations assume a constant CO-to-H_2 conversion factor $X_{CO} = 2 \times 10^{20}$ cm^{-2} (K km s^{-1})$^{-1}$, which is the X_{CO} value for solar neighborhood metallicity. Equation 2.3 also includes corrections for the J=2–1 to J=1–0 line ratio. The middle column of Figure 3 displays the Σ_{H_2} maps.

2.4 Extraction of kinematically cold and warm HI components

2.4.1 Overview of Gaussian decomposition approaches

As discussed in Section 1.1.3, several methods exist for decomposing HI emission spectra into multiple Gaussian components that can be associated with kinematically distinct gas populations. These methods differ primarily in how they handle the inherent degeneracies in multi-Gaussian fitting and how they incorporate spatial information.

Pixel-by-pixel decomposition methods, such as BAYGAUD (Oh et al., 2019), fit Gaussian models to each line-of-sight velocity profile independently. BAYGAUD uses Bayesian Markov Chain Monte Carlo techniques with nested sampling to estimate model parameters and uses Bayes factor statistics to determine the optimal number of Gaussian components. While this approach provides robust parameter

[3]https://iram-institute.org/observatories/30-meter-telescope/

[4]https://oms.iram.fr/oms/?dms=viewobject/_=-O-260678&pageId=4

estimation and uncertainty quantification, it treats each spatial pixel independently, potentially missing coherent structures that extend across multiple pixels. Another pixel-by-pixel approach, GaussPy (Lindner et al., 2015), uses an autonomous Gaussian decomposition (AGD) algorithm based on derivative spectroscopy to determine the number of components and their initial parameters, and has been applied specifically to separate narrow, CNM-like components from broad, WNM-like components in HI emission and absorption spectra (Murray et al., 2017, 2018). Its successor, GaussPy+ (Riener et al., 2019), improves the noise handling and fitting routine of the original algorithm and introduces an optional spatial refitting step, in which flagged or poorly constrained solutions are refit using the results from neighbouring pixels. This has also been used for cold atomic gas extraction from HI emission (Syed et al., 2023). Unlike BAYGAUD, GaussPy+ can therefore incorporate a limited degree of spatial coherence, though this is applied as a post-processing correction rather than as an intrinsic constraint of the fit itself.

An alternative approach, the super-profile method of Ianjamasimanana et al. (2012), addresses the signal-to-noise limitations of pixel-by-pixel-based approaches by stacking velocity profiles across entire galaxies or large regions. While effective for deriving global properties, this method sacrifices spatial information and cannot provide maps of individual phase components.

For this study, we use the spatially coherent Gaussian decomposition methodology implemented in ROHSA (Regularised Optimisation for Hyper-Spectral Analysis; Marchal et al. 2019) to separate the HI emission in our galaxies into narrow and broad HI components. ROHSA incorporates spatial coherence as a constraint in the decomposition, producing maps of kinematically distinct components that are physically connected across neighbouring pixels.

2.4.2 The ROHSA algorithm

ROHSA was developed specifically to address the challenge of extracting the multiphase structure of the neutral ISM from 21 cm emission data cubes (Marchal et al., 2019). The algorithm decomposes hyper-spectral data into a sum of spatially coherent Gaussian components, taking advantage of the physical expectation that individual gas structures should vary smoothly across adjacent lines of sight.

ROHSA models the observed brightness temperature spectrum $T_B(x,y,v)$ at each spatial position (x,y) and velocity v as a sum of N Gaussian functions:

$$T_B(x,y,v) = \sum_{n=1}^{N} A_n(x,y) \exp\left(-\frac{(v-\mu_n(x,y))^2}{2\sigma_n(x,y)^2}\right) \tag{2.5}$$

where for each Gaussian component n, $A_n(x,y)$ is the amplitude, $\mu_n(x,y)$ is the centroid velocity, and $\sigma_n(x,y)$ is the velocity dispersion. Unlike pixel-by-pixel methods, ROHSA constrains these parameters to vary smoothly across the spatial dimensions.

The algorithm minimises a regularised cost function that combines a data fidelity term with spatial regularisation terms:

$$\mathcal{J}(\theta) = \sum_{x,y,v} \frac{(T_B^{\rm obs}(x,y,v) - T_B^{\rm model}(x,y,v))^2}{\sigma_{\rm noise}(x,y)^2} + \mathcal{R}(\theta) \tag{2.6}$$

where θ represents the full set of Gaussian parameters, and $\mathcal{R}(\theta)$ is the regularisation term that enforces spatial smoothness. The regularisation term takes the form:

$$\mathcal{R}(\theta) = \lambda_A \sum_n \|\nabla A_n\|^2 + \lambda_\mu \sum_n \|\nabla \mu_n\|^2 + \lambda_\sigma \sum_n \|\nabla \sigma_n\|^2 + \lambda''_\sigma \sum_n (\sigma_n - \bar{\sigma}_n)^2 \tag{2.7}$$

The first three terms penalise spatial gradients in the amplitude, centroid velocity, and velocity dispersion fields, respectively, encouraging smooth variations across the map. The hyperparameters λ_A, λ_μ, and λ_σ control the strength of this spatial smoothing. The fourth term, controlled by λ''_σ, encourages the velocity dispersions of each Gaussian component to cluster around characteristic values $\bar{\sigma}_n$, promoting the identification of distinct kinematic phases.

ROHSA implements a multi-resolution optimisation strategy. Rather than attempting to fit the full-resolution data cube directly, the algorithm proceeds through a hierarchical series of spatial scales, from coarse to fine resolution:

1. The data cube is spatially averaged to a coarse grid.
2. Gaussian parameters are estimated on this smoothed version.
3. The solution is interpolated to a finer grid.
4. Parameters are refined at the new resolution.
5. Steps 3–4 are repeated until the original resolution is reached.

This multi-resolution approach improves convergence by first capturing large-scale structures before refining small-scale details. The optimisation at each resolution level is performed using the L-BFGS-B algorithm (Byrd et al., 1995), a limited-memory quasi-Newton method suitable for large-scale bound-constrained optimisation.

2.4.3 Advantages of ROHSA for phase separation

ROHSA offers several advantages over alternative decomposition methods for the purpose of separating kinematically cold and warm HI components:

1. **Spatial coherence**: By incorporating regularisation, ROHSA produces Gaussian components that are spatially coherent, reflecting the physical expectation that individual gas structures extend across multiple resolution elements. This is particularly important for identifying extended cold HI structures that may be associated with molecular cloud formation regions.
2. **Phase separation capability**: The λ''_σ regularisation term explicitly encourages the identification of components with distinct characteristic velocity dispersions, facilitating the separation of kinematically narrow (cold) and broad (warm) components without requiring prior assumptions about the number of phases.
3. **Noise suppression**: The spatial regularisation naturally suppresses noise-driven fluctuations in the fitted parameters, improving the reliability of the decomposition in regions of lower signal-to-noise ratio.
4. **Scalability**: The multi-resolution approach and efficient optimisation algorithm allow ROHSA to process large data cubes in reasonable computation times.

Marchal et al. (2019) tested ROHSA on synthetic 21 cm observations computed from numerical simulations of thermally bistable turbulence, demonstrating that the algorithm successfully recovers the multiphase nature of the HI and that the power spectra of column density and centroid velocity for each phase are well reproduced. Application to real observations of a high Galactic latitude field from the GHIGLS survey produced a picture of multiphase HI with isolated, filamentary narrow structures ($\sigma \sim 1$–2 km s^{-1}) and broader, diffuse components ($\sigma \sim 4$–10 km s^{-1}).

2.4.4 Implementation and parameter selection

ROHSA is implemented in Fortran 90 and is publicly available[5]. A Python interface (ROHSApy) facilitates running the code and analysing the output. The algorithm requires several user-specified inputs:

- The number of Gaussian components N: This should be chosen high enough to adequately encode the spectral complexity but not so high as to overfit the data.

[5] https://github.com/antoinemarchal/ROHSA

- The hyperparameters (λ_A, λ_μ, λ_σ, λ''_σ): These control the degree of spatial smoothing and phase clustering. Higher values produce smoother maps but may suppress real small-scale structure; values that are too low may result in noise-dominated solutions.
- A 2D noise map: needed for better weighting in the cost function, but is not mandatory

These parameters must be determined empirically for each galaxy, as optimal values depend on the data quality, spatial resolution, and complexity of the velocity field. For this study, parameters were tuned through an iterative process, examining the residuals and the physical plausibility of the resulting decomposition.

As an example, Figure 4 shows the total, kinematically cold, and kinematically warm HI surface density maps for IC 2574. The final fits use $\lambda_A = 0$, $\lambda_\mu = 1$, $\lambda_\sigma = 1$ and $\lambda''_\sigma = 0$, with two or three Gaussian components per galaxy. Removing the global width penalty allows the fitted dispersions to change across the disc, including with radius. Figure 4 uses the final fit with two Gaussians for IC 2574. The adopted parameters are presented in Appendix A.

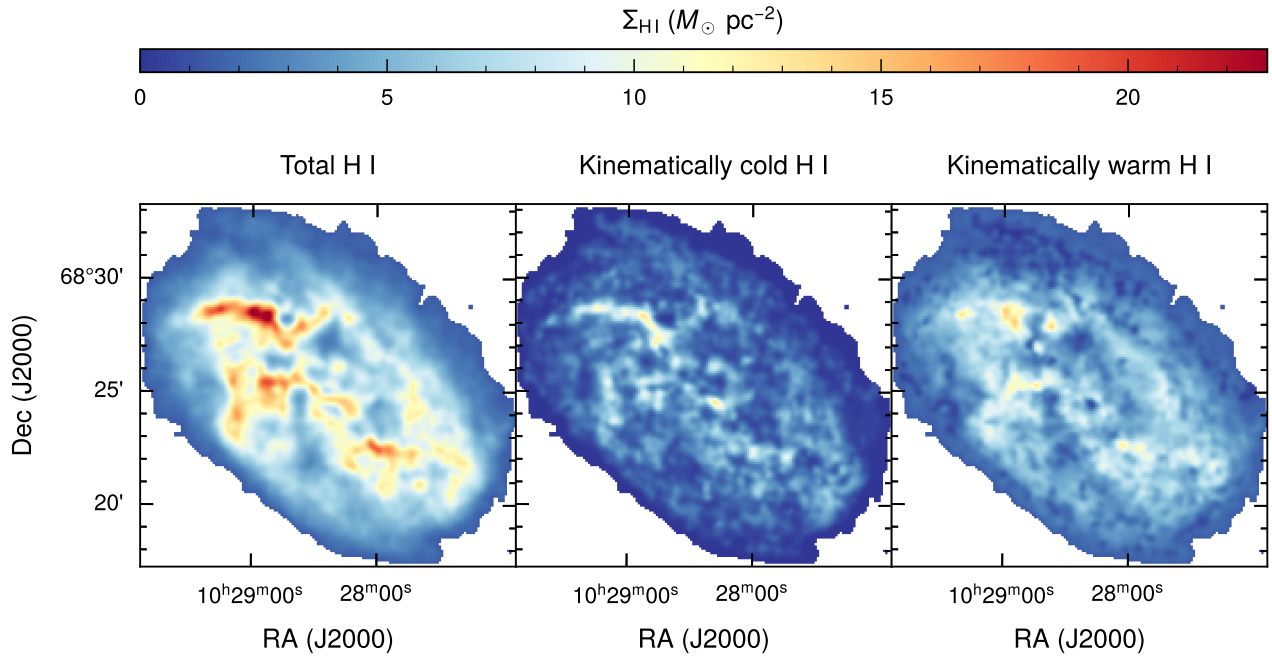


Figure 4: Total (left), kinematically cold (middle), and kinematically warm (right) HI surface density maps of IC 2574 from the ROHSA decomposition. The total map is the sum of the cold and warm components. All panels use the same colour scale and mask. The shared colourbar shows the surface density in $M_\odot$ pc^{-2}.

2.4.5 Classification of kinematically cold and warm components

Following the Gaussian decomposition, individual components were classified as kinematically cold or warm based on their velocity dispersions. Guided by the observational and theoretical literature (Heiles & Troland, 2003a; Ianjamasimanana et al., 2012; Park et al., 2025), the threshold is measured locally from the fitted widths. At each position, we calculate flux-weighted mean widths for the narrowest and widest Gaussians within a three-beam neighbourhood and take their geometric mean as the dividing width. The cut is bounded below by the resolvable dispersion, $1.274\Delta v$, where Δv is the channel spacing. Components below this local cut are assigned to the narrow map and the others to the broad map. The width distributions do not show a clear separation into two modes, so this cut defines a kinematic partition rather than detecting two thermal phases.

Observed linewidths include thermal and nonthermal broadening. The terms kinematically cold and kinematically warm describe the narrower and broader components, rather than directly identifying them with the thermal CNM and WNM. The narrow-component fraction, f_n, is the fraction of integrated HI flux assigned to the narrow component. This fraction also depends on the assumed number of Gaussian components. BAYGAUD provides a comparison because it selects the component number separately for each sight line.

2.4.6 Requirements and limitations

The ROHSA algorithm operates under the assumption that the data are optically thin, meaning that most radiation passes through the gas without significant absorption or scattering. This assumption is generally valid for face-on or moderately inclined galaxies, but becomes problematic for edge-on systems where line-of-sight path lengths through the disc are long. Consequently, the sample was restricted to galaxies with inclinations $i < 70^\circ$.

Computation time depends on several factors, primarily the number of Gaussian components N, the spatial dimensions of the data cube (in pixels), and the number of velocity channels.

2.5 Source finding and masking with SoFiA

The Source Finding Application (SoFiA; Serra et al. 2015)[6] was used to generate and mask the HI moment 0 maps from the HI, narrow HI and broad HI data cubes outputted by ROHSA (see Section

[6]https://gitlab.com/SoFiA-Admin/SoFiA-2/

2.4). SoFiA is an open-source program designed for automated detection and characterisation of HI sources in extragalactic spectral line data cubes.

SoFiA uses the position of the emission on the sky and its distribution across velocity channels to identify sources. It provides several detection methods, including methods that search for emission over different spatial and velocity scales. This helps detect both compact sources and faint, extended emission. The detections are recorded in a binary mask, which assigns a value of one to detected voxels, the pixels of a three-dimensional cube, and zero elsewhere. The mask therefore defines where the detected emission lies on the sky and across velocity channels.

An appropriate detection threshold needs to be chosen to distinguish emission from noise, taking into account the sensitivity of the observations and the emission being studied. SoFiA groups neighbouring detected voxels into individual sources. Minimum size requirements can help remove isolated noise peaks. After sources have been identified, an optional reliability parameter can be set to retain only those with a sufficiently high estimated probability of being real.

SoFiA produces integrated intensity maps (moment 0), intensity-weighted velocity maps (moment 1), velocity dispersion maps (moment 2), integrated spectra, position–velocity diagrams, and data cubes for each detected source.

2.6 narrow HI and CO comparison

To test whether the narrow HI component is preferentially associated with CO emission (discussed in section 3.1), we compare its fractional contribution in equal areas of clearly CO-bright and CO-faint emission at similar galactocentric radii in eight galaxies where CO emission is sufficiently bright for this comparison. We start from the native HI and CO maps and convolve them to the same beam size. The resulting spatial resolutions range from 234 to 721 pc.

2.7 Kennicutt–Schmidt relation Fitting methodology

To test which of the kinematically cold and warm HI components is more closely linked to star formation, we fit the Kennicutt–Schmidt relation for each gas component and compare the slopes and correlation strengths, using total HI and CO-traced H_2 as references. Following Bigiel et al. (2008), we compare the slopes of the Kennicutt–Schmidt relations for the different gas components. In logarithmic

space, the Kennicutt–Schmidt relation in Equation 1.3 becomes

$$\log_{10}\Sigma_{\rm SFR} = a + N\log_{10}\Sigma_{\rm gas}, \tag{2.8}$$

where $a = \log_{10}A$ is the intercept and N is the power-law index.

Bigiel et al. (2008) used the ordinary least-squares (OLS) bisector to fit their Kennicutt–Schmidt relations. A single OLS regression minimises the vertical or horizontal residuals, depending on which variable is chosen as dependent. The bisector combines these two regressions and treats the axes symmetrically. Several atomic-gas distributions in our sample have considerable scatter or are nearly vertical. We therefore adopt orthogonal regression, which minimises the distances perpendicular to the fitted line and estimates the direction of the distribution directly.

Since neighbouring pixels within a beam contain correlated measurements of the same emission, we sample the maps at positions separated by at least one beam width, both horizontally and vertically. Before doing so, we first match the gas and SFR maps to the same spatial resolution and pixel grid. At each selected position, we use the gas and SFR values from the corresponding pixel. We fit the Kennicutt–Schmidt relation to these beam-sampled measurements, shown as blue points in Figure 5. The grey shading shows all positive pixel values across the mapped field.

Each beam-sampled measurement in the Kennicutt–Schmidt relation has coordinates (x_i, y_i), where $x_i = \log_{10}\Sigma_{{\rm gas},i}$ and $y_i = \log_{10}\Sigma_{{\rm SFR},i}$, for $i = 1,\ldots,n$. Here, n is the number of beam-sampled measurements. We use the orthogonal-regression criterion (Pearson, 1901; Isobe et al., 1990), with weights defined below. The fitted line minimises the weighted sum of squared perpendicular distances:

$$\mathcal{Q}(a,N) = \sum_{i=1}^{n} w_i d_{\perp,i}^2 = \sum_{i=1}^{n} w_i \frac{(y_i - a - Nx_i)^2}{1+N^2}, \tag{2.9}$$

where N is the slope of the Kennicutt–Schmidt relation in Equation 2.8, a is the intercept and w_i is the weight assigned to measurement i. The distance $d_{\perp,i}$ is the shortest distance from (x_i, y_i) to the fitted straight line $y = a + Nx$ on the plot of $\log_{10}\Sigma_{\rm SFR}$ against $\log_{10}\Sigma_{\rm gas}$.

To reduce the influence of sparsely distributed data points, we give greater weight to points in densely populated parts of the gas–SFR plot and less weight to points in sparse regions. We estimate the local point density using a two-dimensional Gaussian kernel (Scott, 1992). For each blue point i, we define its position in the logarithmic gas–SFR plot as

$$\mathbf{z}_i = \begin{pmatrix} x_i \\ y_i \end{pmatrix}. \tag{2.10}$$

We assign this point a weight proportional to the density of neighbouring blue points in the gas–SFR plot:

$$w_i = \sum_{\substack{j=1 \\ j\neq i}}^{n} \exp\left[-\frac{1}{2}(\mathbf{z}_i - \mathbf{z}_j)^{\mathrm{T}}\mathbf{H}^{-1}(\mathbf{z}_i - \mathbf{z}_j)\right], \tag{2.11}$$

where the sum includes all other blue points. The matrix $\mathbf{H}$ sets the width and shape of the Gaussian neighbourhood. We use Scott's bandwidth rule (Scott, 1992), giving $\mathbf{H} = n^{-1/3}\mathbf{S}$, where n is the number of blue points and $\mathbf{S}$ is the covariance matrix of their logarithmic coordinates. Nearby points contribute strongly to the sum, while distant points contribute very little. A point with more nearby neighbours therefore receives a larger weight. These weights describe the concentration of measurements in the plot; they are not measurement-error weights.

We use the Pearson correlation coefficient r to quantify the strength and direction of the correlation in each Kennicutt–Schmidt relation.

To estimate the uncertainties in the fitted parameters, we generate 2000 bootstrap samples by resampling the paired gas and SFR measurements with replacement. We fit the relation to each resampled data set. We use the 16th and 84th percentiles of the fitted values from these samples to define the lower and upper limits of the 68% uncertainty interval. These intervals describe sampling uncertainty and exclude systematic uncertainties in the CO-to-H_2 conversion and SFR calibration. For nearly vertical relations where the slope is poorly constrained, we report the angle θ of the fitted line to the horizontal axis. The dashed lines in Figure 5 show the fitted relations. Table 3 gives the fitted values and their uncertainties.

2.8 Galaxy sample

For this study, the classification scheme of Leroy et al. (2008) was adopted to distinguish between dwarf and spiral galaxies: dwarf galaxies are defined as having stellar masses $\log(M_*/M_\odot) \lesssim 10.0$ and absolute B-band magnitudes $M_B \gtrsim -20$ mag. Galaxies not meeting these criteria are classified as spirals.

The final sample consists of 16 galaxies: nine dwarfs and seven spirals. These were selected from the intersection of THINGS, GALEX NGS and SINGS, ensuring the availability of all required data products. All the low-mass galaxies except NGC 0925, NGC 2403, NGC 2976 and NGC 4214 have no CO data, so they are assumed to be HI dominated.

Four galaxies from the initial sample were excluded. For Ho I, M81 dwA, and NGC 5055, none of the tested ROHSA parameter settings met the adopted acceptance criteria. NGC 5236 was excluded

because strong negative features in its HI cube affected the existing decomposition results.

All selected galaxies have inclinations $i < 70°$. This criterion was imposed for several reasons: (1) highly inclined galaxies yield fewer independent data points after deprojection, complicating interpretation; (2) beam elongation along the minor axis introduces systematic uncertainties; and (3) the optically thin assumption required by ROHSA becomes increasingly violated at high inclinations.

The final HI decompositions use each galaxy's native spatial resolution. For each galaxy, the gas and SFR maps were matched to a common spatial resolution before comparison. Comparisons between galaxies also require smoothing the maps to a common physical resolution so that the same spatial scale is studied in each galaxy. Table 2 lists the sample galaxies and their properties.

Table 2: ***Properties of the final galaxy sample.*** *D is the distance, i the inclination, r_{25} the optical radius, PA the position angle, M_* the stellar mass, M_B the absolute B-band magnitude, SFR the star formation rate, metallicity is given as $12+\log(\mathrm{O/H})$, ℓ_{HI} is the native linear resolution of the HI data, and ℓ_{dec} is the resolution used for decomposition.*

Galaxy	D [Mpc]	i [°]	r_{25} [kpc]	PA [°]	Hubble Type	Morph. Type	$\log M_*$ [$M_\odot$]	M_B [mag]	SFR [$M_\odot\ yr^{-1}$]	metal. 12+log(O/H)	ℓ_{HI} [pc]	ℓ_{dec} [pc]
1	2	3	4	5	6	7	8	9	10	11	12	13
Dwarf galaxies												
DDO 53	3.6	31	0.4	132	Irr	9.9	7.0[b]	−13.4	0.008[c]	7.77	205	205
DDO 154	4.3	66	1.2	230	Irr	9.9	7.1	−14.4	0.005	7.54	294	294
HO II	3.4	41	3.7	177	Irr	9.9	8.3	−16.9	0.048	7.68	226	226
IC 2574	4.0	53	7.5	56	Irr	8.9	8.7	−18.0	0.070	7.94	248	248
NGC 0925	9.2	66	14.2	287	SBcd	7.0	9.9	−20.0	0.561	8.24	265	265
NGC 2403	3.2	63	7.3	124	SBc	6.0	9.7	−19.4	0.382	8.31	136	136
NGC 2976	3.6	65	3.8	335	Sc	5.2	9.1	−17.8	0.087	8.30	129	129
NGC 4214	2.9	44	2.9	65	Irr	9.8	8.8	−17.4	0.107	8.34	207	207
NGC 7793	3.9	50	5.8	290	Scd	7.4	9.5	−18.7	0.235	8.22	295	295
Spiral galaxies												
NGC 0628	7.3	7	10.4	20	Sc	5.2	10.1	−20	0.807	8.33	420	420
NGC 2903	8.9	65	11.7	204	Sbc	4.0	10.6[a]	−17.8	1.400	9.12	659	659
NGC 3184	11.1	16	11.9	179	SBc	5.9	10.3	−19.9	0.901	8.48	287	287
NGC 3351	10.1	41	10.6	192	SBb	3.1	10.4	−19.7	0.940	8.60	487	487
NGC 4736	4.7	41	5.3	296	Sab	2.3	10.3	−20.0	0.481	8.31	233	233
NGC 5194	8.0	20	9.0	172	SBc	4.0	10.6	−21.1	3.125	8.54	462	462
NGC 6946	5.9	33	9.8	243	SBc	5.9	10.5	−20.9	3.329	8.40	173	173

Notes. *Unless noted, columns 2 – 6 and 8 – 10 are from Leroy et al. (2008). Column 11 is from Ianjamasimanana (2014). Column 12 gives the physical size of the native HI beam major axis, calculated from the original cube headers using the distances in column 2. Column 13 gives the physical resolution used for the HI decomposition. Column 7 is from Makarov et al. (2014).* [a] *– from Leroy et al. (2021),* [b] *– from Sylos Labini et al. (2025),* [c] *– from Walter et al. (2008)*

2.9 Open science and reproducibility

To make this thesis fully reproducible, we make all analysis scripts available on GitHub at `https://github.com/AnjiaRandia/Thesis_M2`. We use Snakemake to organise our processing scripts into a workflow that can be launched with a single command. Snakemake is a workflow management system that automates the execution of analysis steps in the order required by their inputs and outputs (Mölder et al., 2021). Separate configuration files record the adopted analysis settings, while an environment file lists the software dependencies.

Sharing the analysis scripts allows other researchers to reproduce our analysis and adapt it to their own data. This approach aligns with the FAIR principles, which aim to make research products Findable, Accessible, Interoperable and Reusable (Wilkinson et al., 2016).

These principles also guide the open science approach promoted by the SKA Observatory (SKAO), which encourages the sharing of data, software and scientific workflows (SKA Observatory, 2024). Its network of SKA Regional Centres is being developed to provide access to data, computing resources and analysis tools, allowing researchers to collaborate and reproduce one another's analyses. The sharing of our analysis workflow follows the same aim of making our methods fully accessible to others.

Chapter 3: Results and discussions

In this chapter, we examine how the kinematically cold and warm HI components are distributed in nearby galaxies and how closely each component is linked to molecular gas and star formation.

3.1 Spatial distribution of the kinematically cold and warm components

Figure 5 shows the total, kinematically cold and kinematically warm HI surface-density maps for the individual galaxies. Both components extend across much of the mapped atomic discs. Both components are present in clumps and ridges within the extended HI emission.

The kinematically cold component contributes a median of about 45% of the mapped HI mass, while the warm component contributes the remaining 55%. Within the optical disc ($r < r_{25}$), the warm component dominates the HI budget, although the cold component remains a substantial part of the atomic gas.

The cold component has a more filamentary appearance than the warm component, with thin, elongated structures standing out against the more diffuse atomic emission. These elongated structures raise the question of whether the cold component is preferentially found in spiral arms. Such a preference could arise because compression in spiral arms can increase the gas density and promote cooling (Kim et al., 2008). After tracing the spiral arms in nine spiral galaxies with visible arm structure in our sample following the methodology used by Foyle et al. (2010), which used the 3.6 μm emission to trace stellar arms, we compared the fraction of the narrow component found in the arms and interarm regions in these galaxies. We found that the kinematically cold component is not consistently higher in the arm regions.

Models of HI thermal balance predict that cold, dense gas can coexist with warm, diffuse gas over a range of thermal pressures (Wolfire et al., 2003). At similar thermal pressure, the cold gas is denser because of its lower temperature. Simulations show that increasing the density of warm atomic gas can allow it to cool and condense into cold structures (Saury, E. et al., 2014). Cold atomic gas can then supply material for molecular-cloud formation (Krumholz et al., 2009a). In the simulations of

Glover & Mac Low (2007), the initially atomic gas cools rapidly, and turbulent compression produces dense regions in which H_2 forms more rapidly. The connection to molecular-cloud formation therefore comes from the high densities that cold atomic gas can reach, together with sufficient shielding. This provides a reason to look for an association between the narrow HI component and CO emission.

We found that both atomic components can be enhanced near CO-bright emission, but the cold fraction is not consistently higher there. The clearest enhancement occurs in NGC 5194, where the area-weighted mean cold fraction is about 45% in the selected CO-bright regions, compared with 29% in nearby CO-faint regions. In NGC 6946, both components increase near bright CO while the cold fraction remains about 30%. The rest of the studied sample provides few suitable matched regions to enable a meaningful comparison between CO-bright and CO-faint emission. We therefore find no consistent evidence that narrow HI is generally more concentrated near CO emission than broad HI.

The absence of a consistent enhancement in the cold fraction does not rule out the fact that cold atomic gas is supplying material for molecular-cloud formation. In atomic-to-molecular transition models, atomic gas remains in shielding layers around molecular interiors, while additional gas can become molecular (Krumholz et al., 2009a; Sternberg et al., 2014). Once hydrogen becomes H_2, it no longer contributes to the HI emission, so an increasing molecular reservoir need not be accompanied by an increasing narrow-HI contribution. Furthermore, H_2 can survive where CO is dissociated (Wolfire et al., 2010). Our CO-faint comparison regions are therefore not necessarily free of H_2. These effects could weaken a spatial contrast between the CO-bright and CO-faint regions. Thus, our comparison does not establish whether the narrow HI component preferentially traces regions where atomic gas becomes molecular. Locating the HI-to-H_2 transition regions in our sample is difficult with our data, since a region may contain H_2 without detectable CO emission (Wolfire et al., 2010).

3.2 Kennicutt–Schmidt relations for the individual gas components

To understand which HI component is more closely linked to star formation, we first examine the Kennicutt–Schmidt relations in individual galaxies at their respective matched spatial resolutions. We then compare the dwarf and spiral samples by combining their measurements at a common physical resolution of 800 pc.

The lower panels of Figure 5 show the Kennicutt–Schmidt relations for total, narrow and broad HI. Where available, H_2 is also shown for comparison. DDO 53 and DDO 154 are not shown because they

provide only one and six beam-spaced measurements, respectively, within the adopted optical-disc boundary at 400 pc.

The atomic-gas relations vary considerably between galaxies (Table 3). Ho II shows weak correlations with all three HI tracers. In IC 2574 and NGC 0925, total HI gives the strongest correlation with the SFR. Of the two separated components, narrow HI correlates more strongly with the SFR than broad HI. In NGC 2403, NGC 2976 and NGC 4214, total and broad HI follow the SFR more closely than narrow HI. NGC 7793 also shows a stronger correlation with total HI than with either separated component. The relative strength of the narrow- and broad-HI correlations with the SFR therefore varies between individual galaxies. This is consistent with Young et al. (2003), who found no simple connection between the incidence of low-dispersion HI and the SFR.

In the spiral galaxies, the narrow-component relations are weak in NGC 0628, NGC 3184 and NGC 6946. NGC 3351 shows a negative narrow-HI correlation, while its broad component shows a weak positive correlation. Both atomic components show positive trends in NGC 2903 and NGC 4736, with the stronger relation for broad HI. NGC 5194 also has a stronger broad than narrow relation, but total HI gives the strongest atomic correlation. The strength and even the sign of the narrow-HI correlation with the SFR differ between these spiral galaxies.

Bigiel et al. (2008) also found that a small range of HI surface density can accompany a much wider range of SFR surface density. They interpreted the saturation of the HI surface density as evidence for a transition to molecular gas, with additional gas found mainly in the molecular phase. Under this interpretation, regions with similar HI surface densities can have different molecular surface densities and SFRs.

These individual-galaxy comparisons use gas and SFR maps matched to the same spatial resolution within each galaxy. The physical size of the beam differs from one galaxy to another. The molecular relations are positive and are clearer than the atomic relations in most spirals. Their fitted slopes range from about 0.86 to 1.20. We therefore recover the approximately linear molecular relation already established by Bigiel et al. (2008), using density-weighted orthogonal regression instead of their OLS-bisector method. Here, the molecular relation provides a reference for assessing whether narrow or broad HI is more closely associated with star formation. Among the lower-mass galaxies, NGC 2976 has a strong molecular correlation and a steeper slope, while NGC 2403 has a moderate, approximately linear relation. NGC 0925 and NGC 4214 have weaker molecular correlations and less well constrained slopes.

To examine the gas–SFR relations across the sample on the same physical scale, we smooth the maps to a common resolution of 800 pc and combine the beam-sampled measurements from the

galaxies. We compare the dwarf and spiral samples separately, as well as the full sample. Figure 6 shows that the combined dwarf and spiral samples behave differently. In the combined dwarf sample, narrow HI correlates more strongly with the SFR ($r = 0.51$) than total HI ($r = 0.45$) or broad HI ($r = 0.23$). In spirals, narrow HI gives the weakest relation ($r = 0.11$), while broad and total HI give $r = 0.37$ and $r = 0.34$. Combining the measurements from all galaxies gives $r = 0.18$, 0.34 and 0.35 for narrow, broad and total HI, respectively.

A possible explanation for the difference between dwarfs and spirals is that line width traces different gas conditions in the two populations. In dwarfs, narrow emission may preferentially trace cooler, less turbulent gas associated with conditions favourable for star formation. In spirals, cold gas near star-forming regions may have its emission broadened by turbulence. Such non-thermal motions can contribute to the line width of cold atomic gas (Saury, E. et al., 2014). Energy input from young stars can help maintain HI turbulence in star-forming discs (Tamburro et al., 2009). Some cold gas in these regions may therefore be assigned to the broad component, which could explain why the broad component shows stronger correlations with the SFR in spirals.

Table 3: ***Resolved gas–SFR fits for the individual galaxies and the combined sample.*** *θ denotes the angle between the fitted line and the horizontal axis and represents the steepness of the relation, with $N = \tan\theta$. For molecular gas, N is the power-law index. The coefficient r is the Pearson correlation coefficient.*

(a) Atomic gas

		Total HI		Narrow HI		Broad HI	
Galaxy	n	θ [°]	r	θ [°]	r	θ [°]	r
Ho II	117	$83.52^{+2.04}_{-2.06}$	0.26	$82.71^{+3.07}_{-2.40}$	0.18	$82.65^{+2.87}_{-2.90}$	0.23
IC 2574	455	$79.94^{+1.09}_{-0.77}$	0.46	$72.12^{+2.08}_{-1.46}$	0.41	$82.87^{+1.28}_{-1.06}$	0.28
NGC 0925	380	$81.87^{+0.79}_{-0.61}$	0.56	$80.96^{+1.45}_{-1.52}$	0.42	$83.23^{+1.33}_{-1.27}$	0.32
NGC 2403	382	$79.46^{+1.11}_{-0.99}$	0.61	$82.90^{+1.11}_{-1.09}$	0.36	$74.00^{+1.19}_{-1.44}$	0.60
NGC 2976	97/96/97	$69.94^{+0.88}_{-1.54}$	0.88	$67.44^{+0.67}_{-6.25}$	0.46	$70.31^{+1.30}_{-1.17}$	0.88
NGC 4214	88	$79.50^{+2.06}_{-2.06}$	0.60	$81.61^{+4.12}_{-4.65}$	0.42	$76.84^{+2.15}_{-2.08}$	0.57
NGC 7793	352	$78.26^{+0.79}_{-0.73}$	0.78	$74.29^{+1.55}_{-1.47}$	0.49	$79.80^{+0.97}_{-1.03}$	0.58
NGC 0628	428	$86.99^{+0.84}_{-0.79}$	0.26	$-87.59^{+0.92}_{-1.19}$	−0.02	$81.44^{+1.40}_{-1.12}$	0.31
NGC 2903	181/177/175	$74.83^{+1.28}_{-1.54}$	0.78	$70.02^{+2.68}_{-2.28}$	0.57	$75.92^{+1.54}_{-1.75}$	0.67
NGC 3184	611	$84.89^{+0.50}_{-0.41}$	0.43	$-89.85^{+0.74}_{-0.81}$	0.04	$81.87^{+0.83}_{-0.80}$	0.44
NGC 3351	314	$87.35^{+0.95}_{-0.51}$	0.01	$-85.74^{+1.33}_{-1.38}$	−0.27	$83.23^{+1.91}_{-1.52}$	0.18
NGC 4736	84	$70.30^{+1.59}_{-0.93}$	0.93	$71.52^{+2.29}_{-1.21}$	0.84	$68.60^{+1.52}_{-1.05}$	0.93
NGC 5194	288	$84.21^{+1.40}_{-1.28}$	0.42	$83.80^{+1.82}_{-1.39}$	0.26	$83.80^{+1.40}_{-1.37}$	0.38
NGC 6946	216	$87.36^{+0.72}_{-0.92}$	0.36	$-89.83^{+2.09}_{-2.76}$	0.07	$86.83^{+1.15}_{-1.60}$	0.34
Dwarfs	769	$83.21^{+0.53}_{-0.63}$	0.45	$80.26^{+0.74}_{-0.91}$	0.51	$85.23^{+0.80}_{-0.87}$	0.23
Spirals	2119	$84.77^{+0.47}_{-0.49}$	0.34	$89.83^{+0.51}_{-0.59}$	0.11	$78.49^{+0.70}_{-0.65}$	0.37
All galaxies	2888	$84.50^{+0.39}_{-0.39}$	0.35	$87.17^{+0.47}_{-0.49}$	0.18	$80.95^{+0.57}_{-0.52}$	0.34

(b) Molecular gas

Galaxy	n	N	r
NGC 0925	257	$2.12^{+0.86}_{-0.72}$	0.25
NGC 2403	259	$0.92^{+0.05}_{-0.12}$	0.58
NGC 2976	86	$1.55^{+0.12}_{-0.14}$	0.85
NGC 4214	66	$2.32^{+0.60}_{-0.95}$	0.40
NGC 0628	387	$1.05^{+0.04}_{-0.04}$	0.73
NGC 2903	146	$1.02^{+0.04}_{-0.06}$	0.84
NGC 3184	522	$1.20^{+0.05}_{-0.07}$	0.64
NGC 3351	264	$1.15^{+0.09}_{-0.10}$	0.71
NGC 4736	79	$1.20^{+0.06}_{-0.06}$	0.91
NGC 5194	276	$0.90^{+0.11}_{-0.01}$	0.72
NGC 6946	212	$0.86^{+0.05}_{-0.05}$	0.84
Dwarfs	507	$1.15^{+0.04}_{-0.17}$	0.35
Spirals	1918	$1.07^{+0.02}_{-0.02}$	0.73
All galaxies	2425	$0.84^{+0.02}_{-0.02}$	0.60

n is the number of fitted beam-spaced measurements.

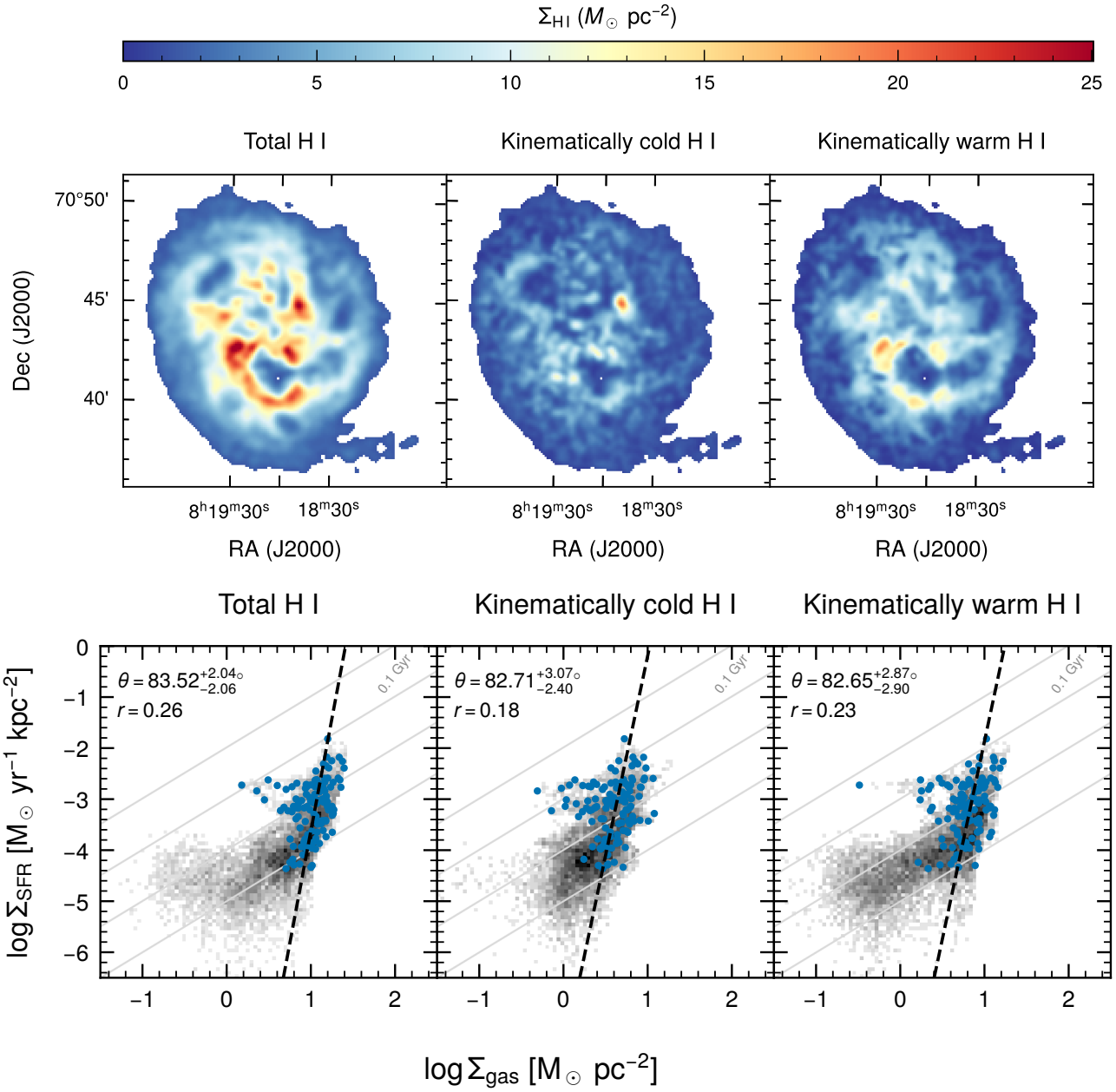


Figure 5: ***Ho II.*** *Top: total, kinematically cold, and kinematically warm HI surface densities. Bottom: Kennicutt–Schmidt relations for total, narrow, and broad HI, and H_2 where included. Grey shading shows all positive mapped pixels. Blue circles show gas and SFR measurements sampled at intervals of at least one beam width. Dashed lines show density-weighted orthogonal fits to these blue points (where available). For the relations between total, narrow or broad HI and the SFR, θ is the angle between the fitted line and the horizontal axis. For the H_2–SFR relation, we report the slope N, or θ if the line is too close to vertical for a finite slope uncertainty to be determined. Intervals are central 68% bootstrap intervals; r is the Pearson coefficient and n_{fit} counts the H_2–SFR measurements used in the fit. Light grey lines mark constant depletion times.*

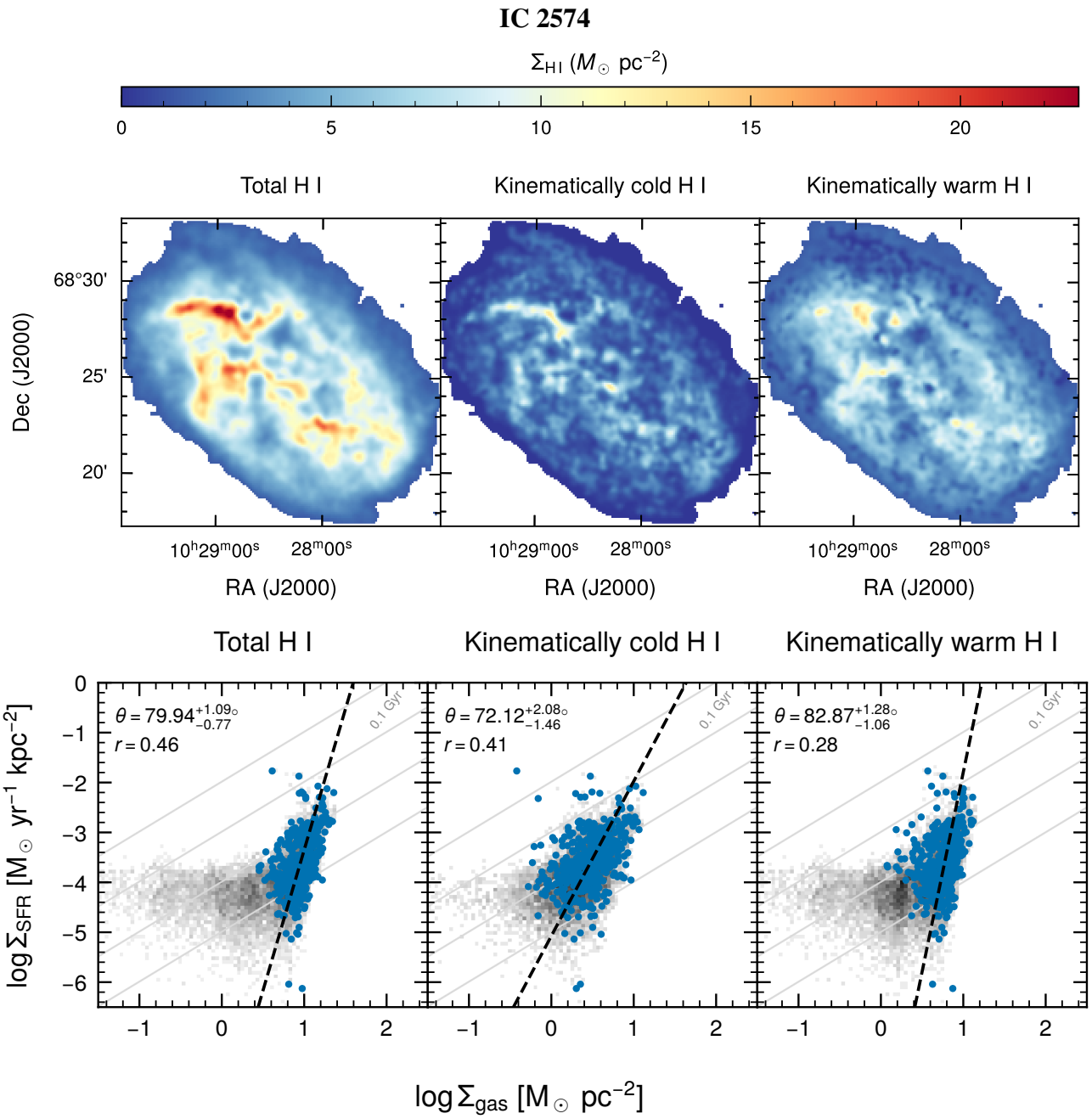


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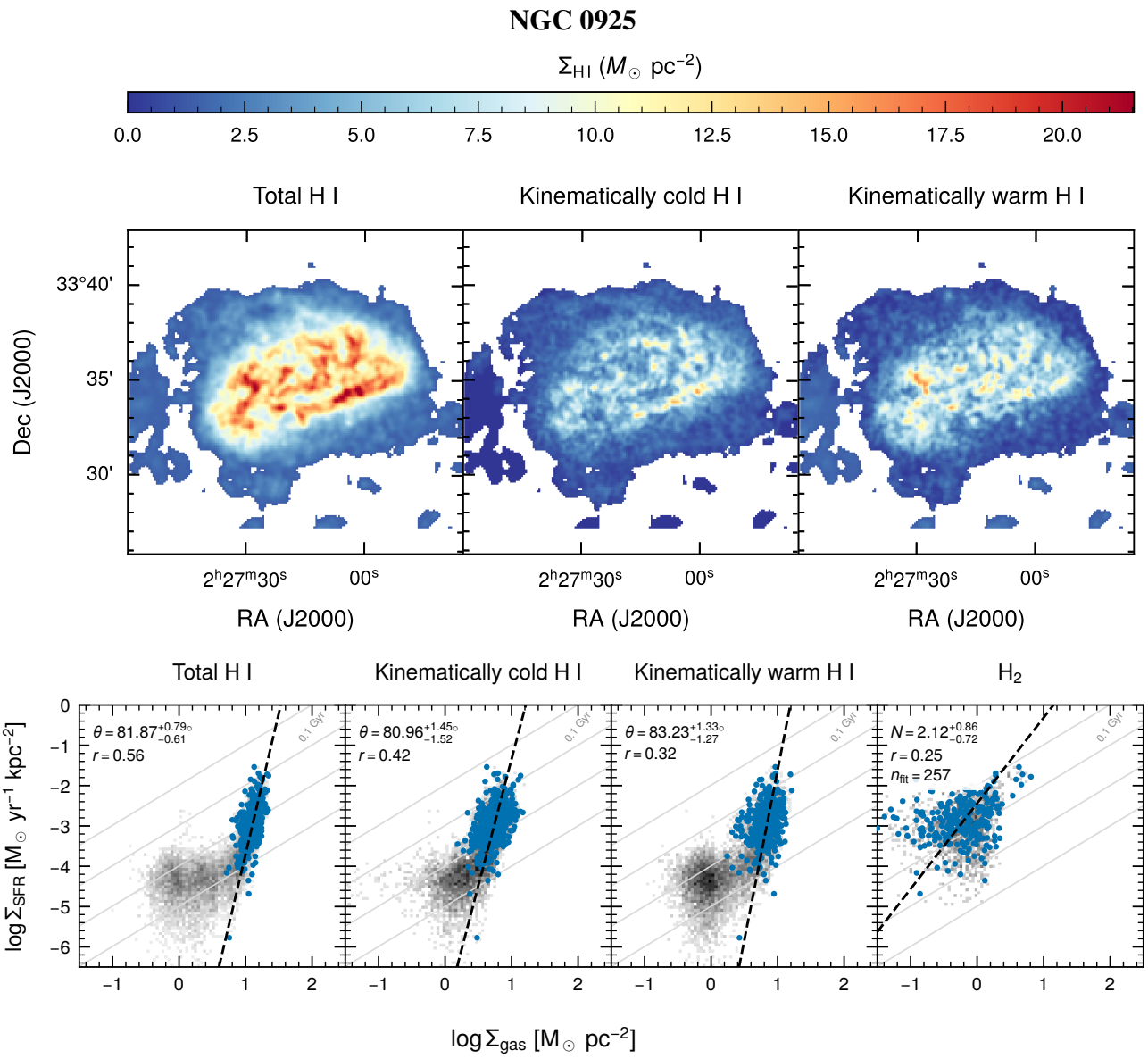


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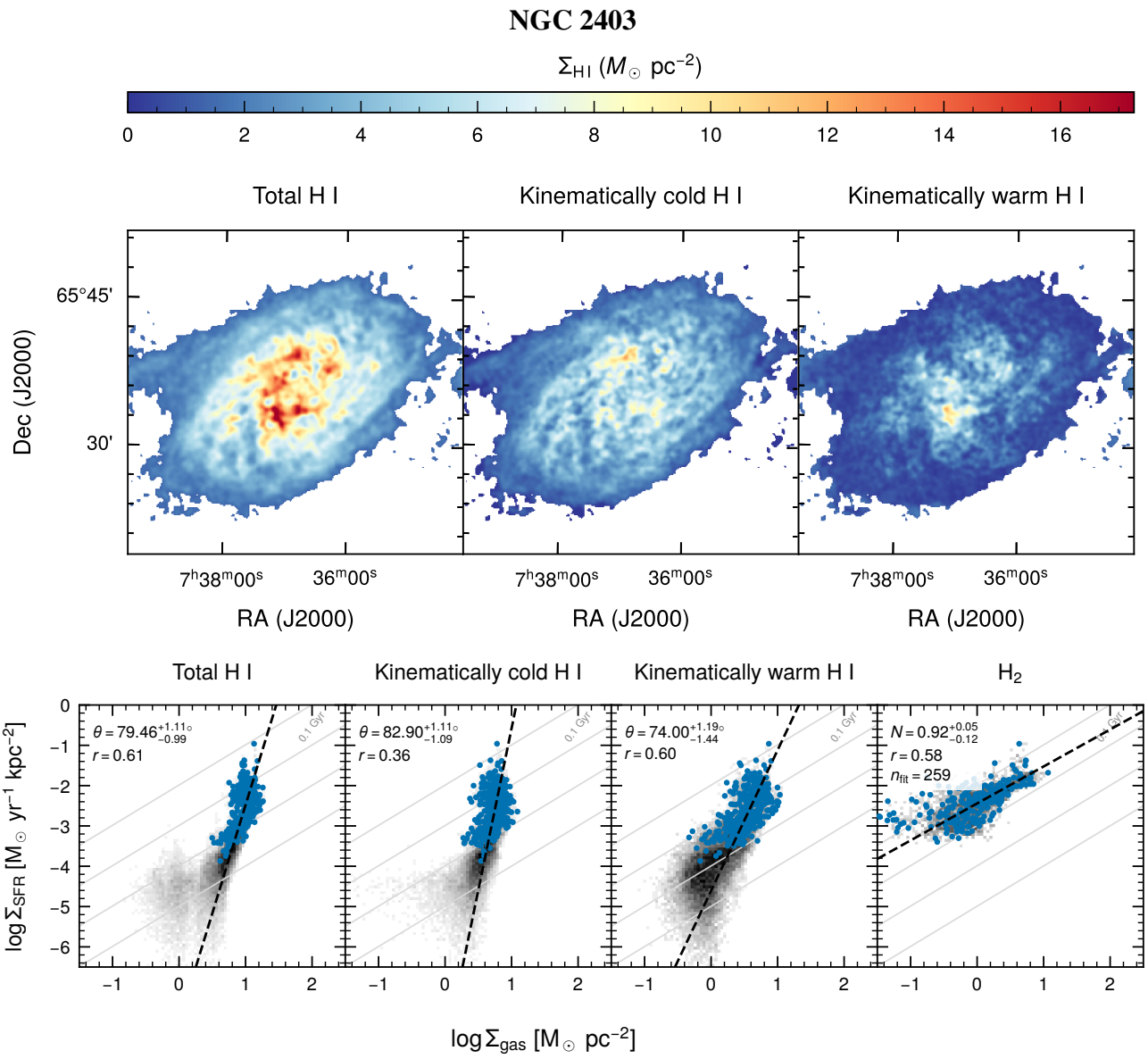


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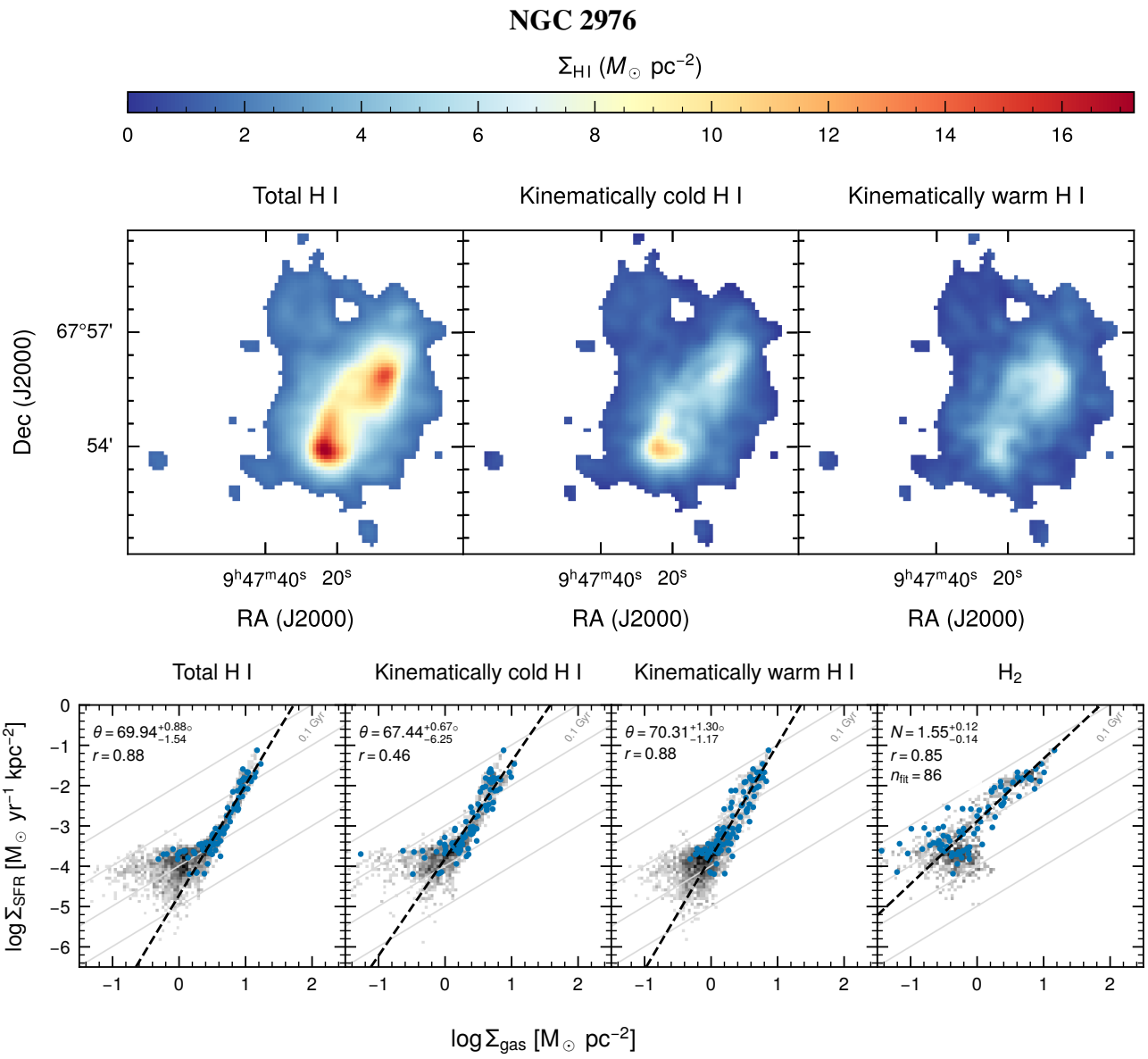


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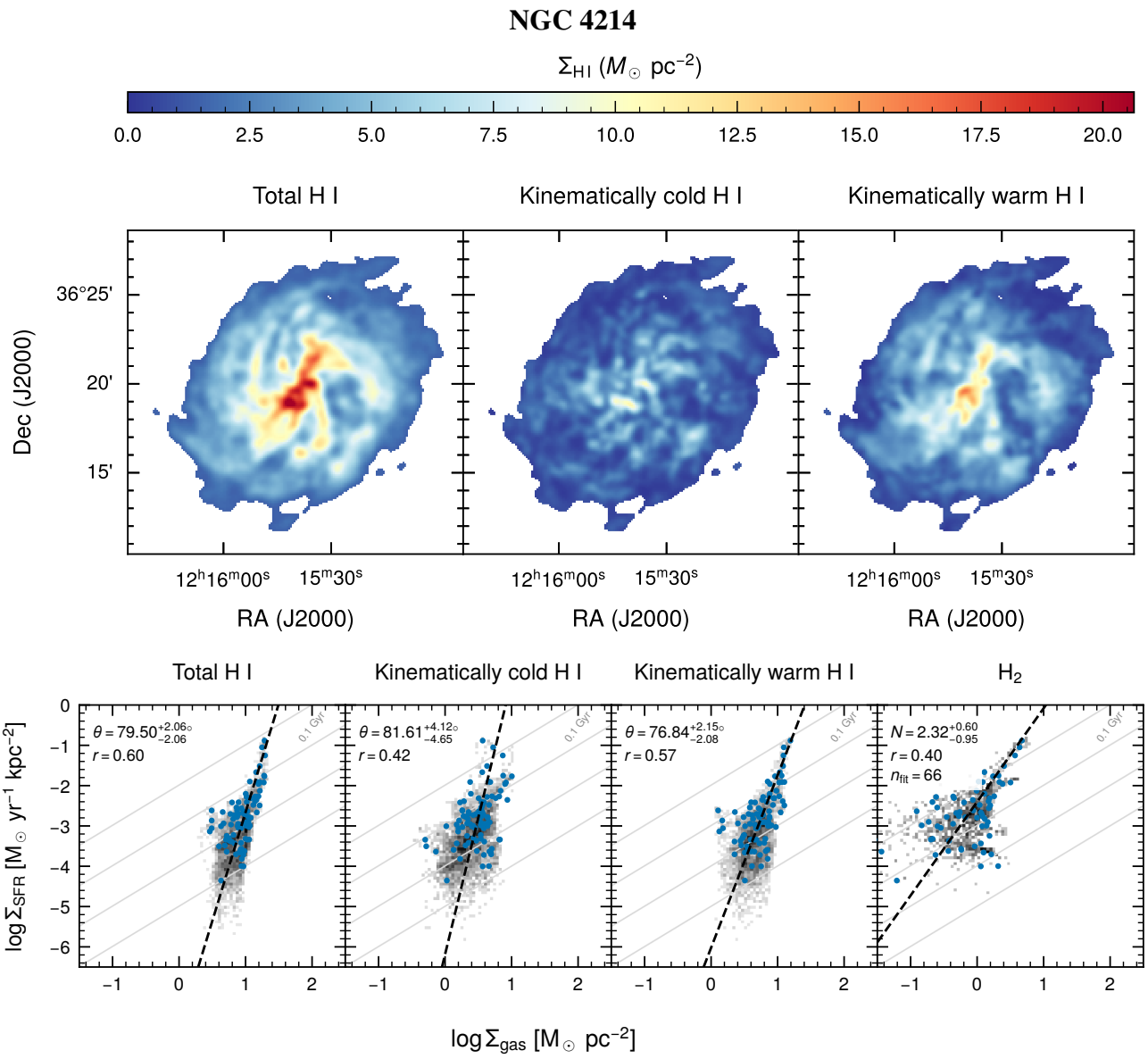


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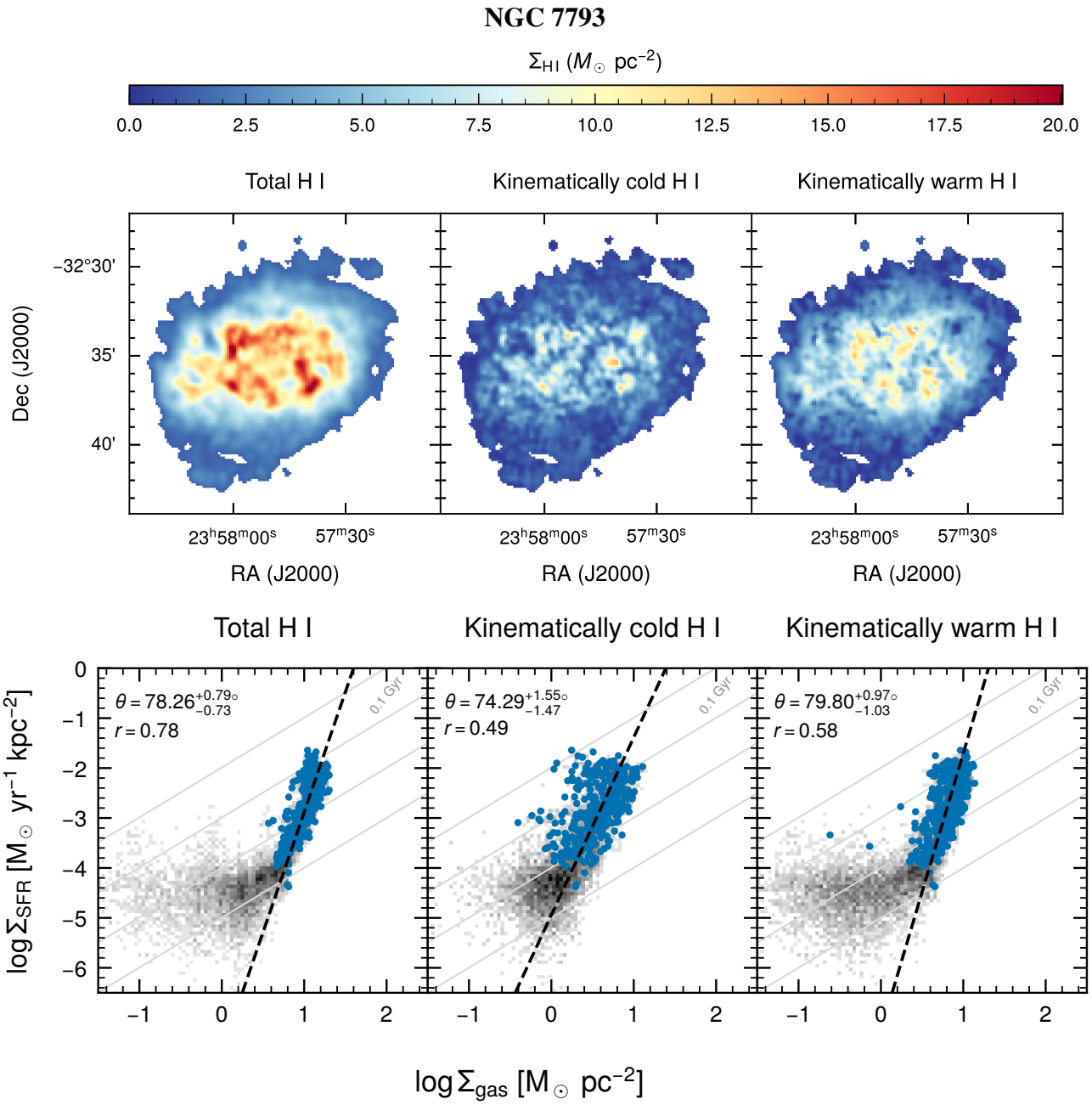


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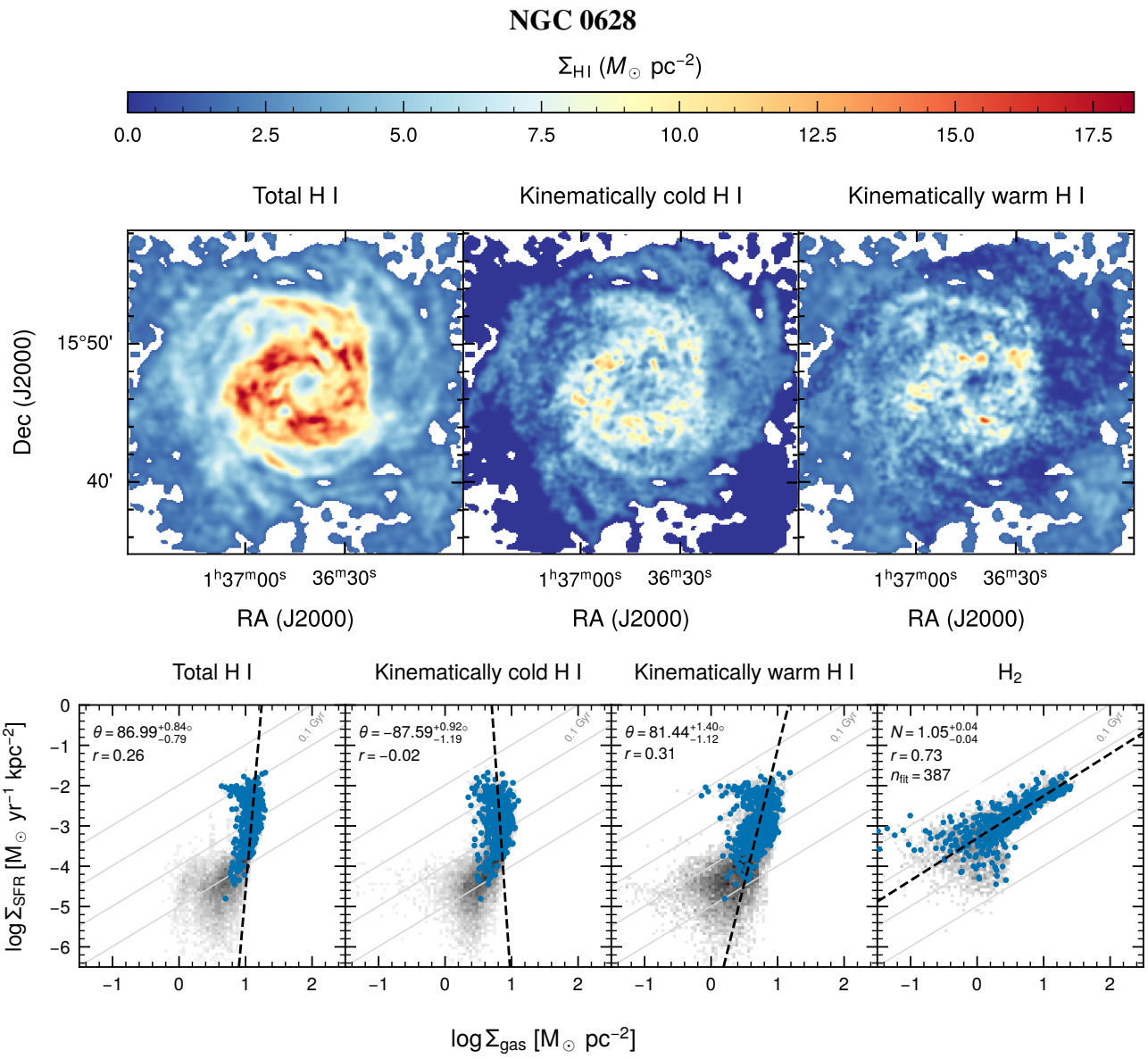


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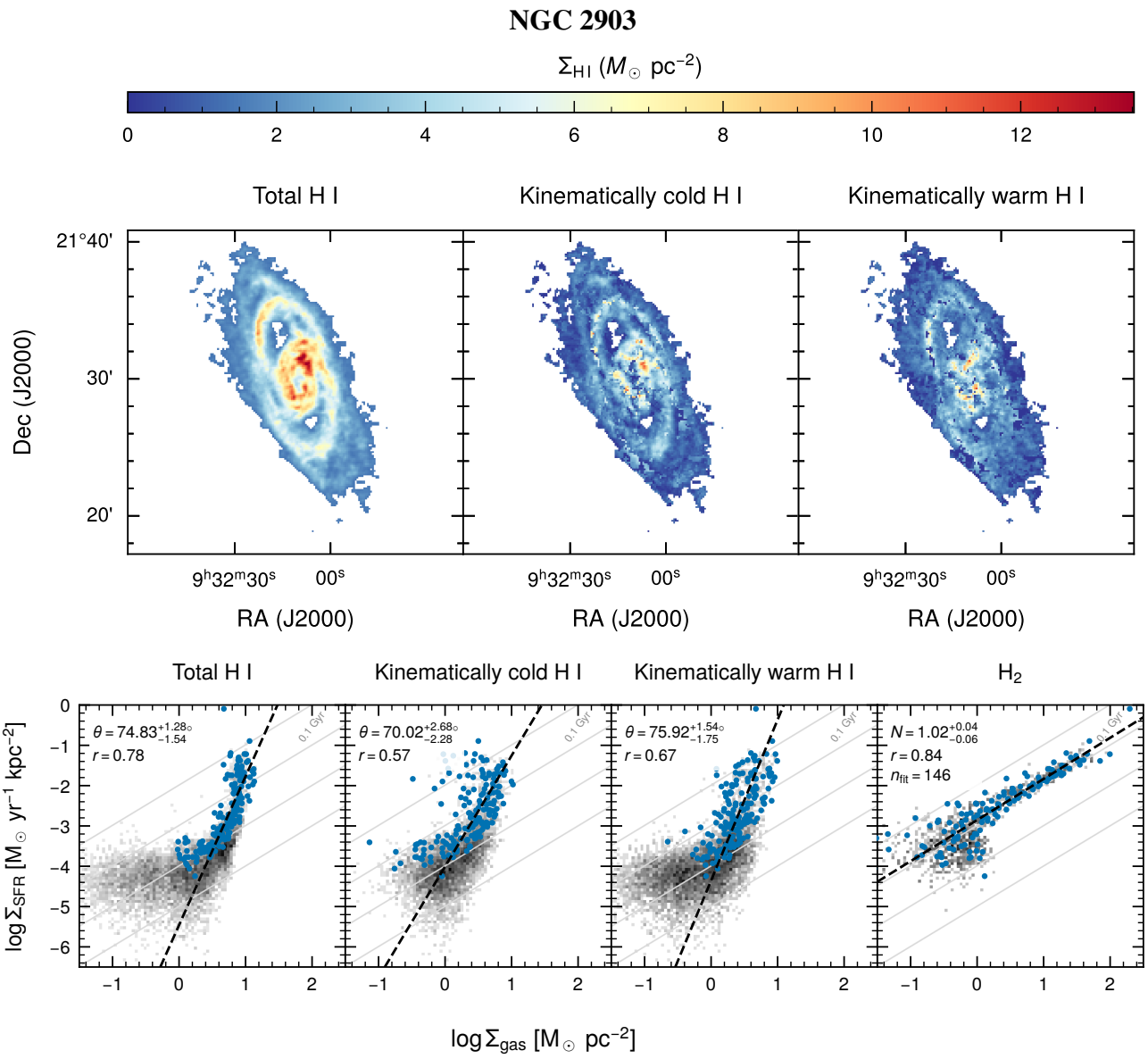


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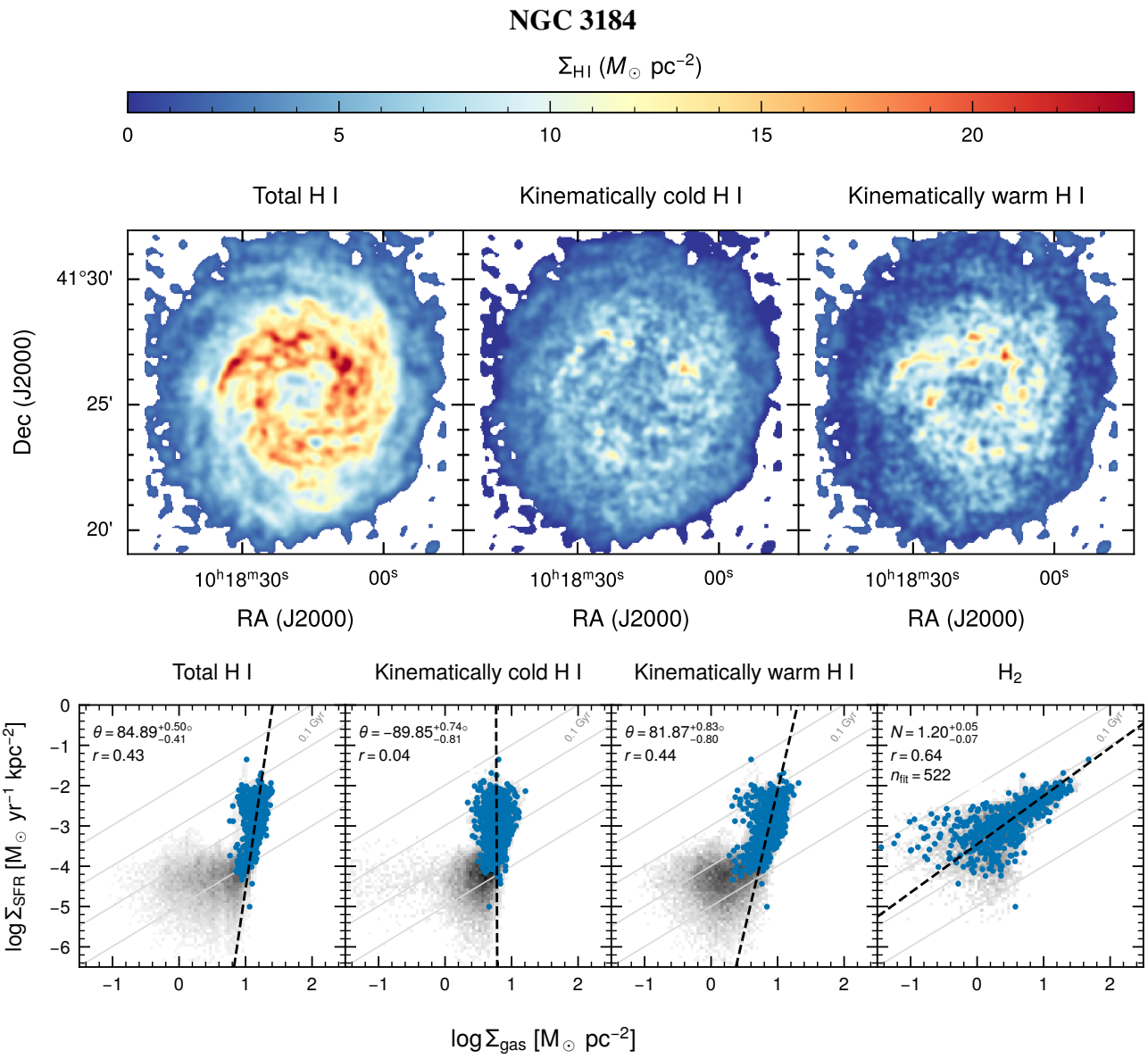


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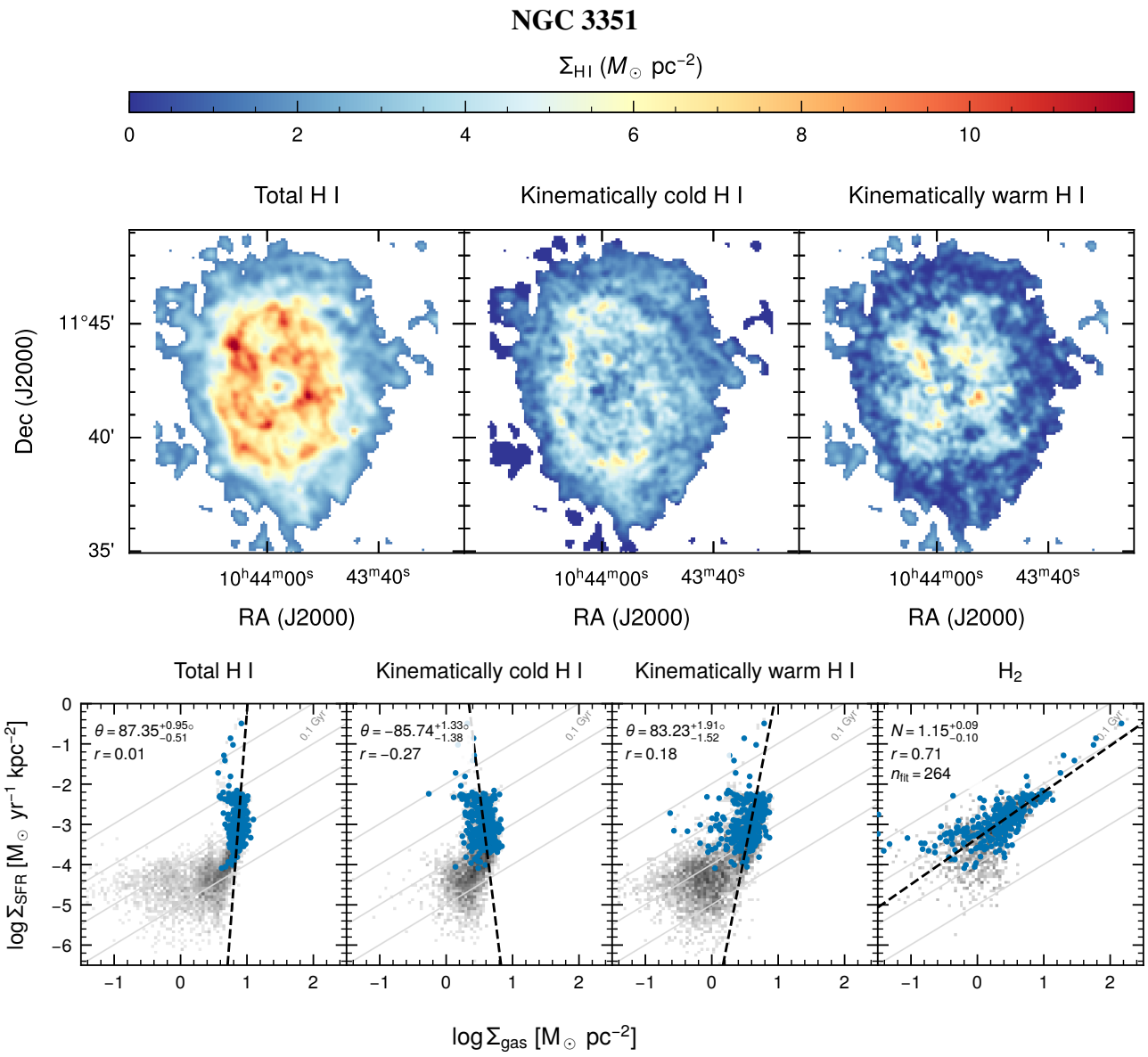


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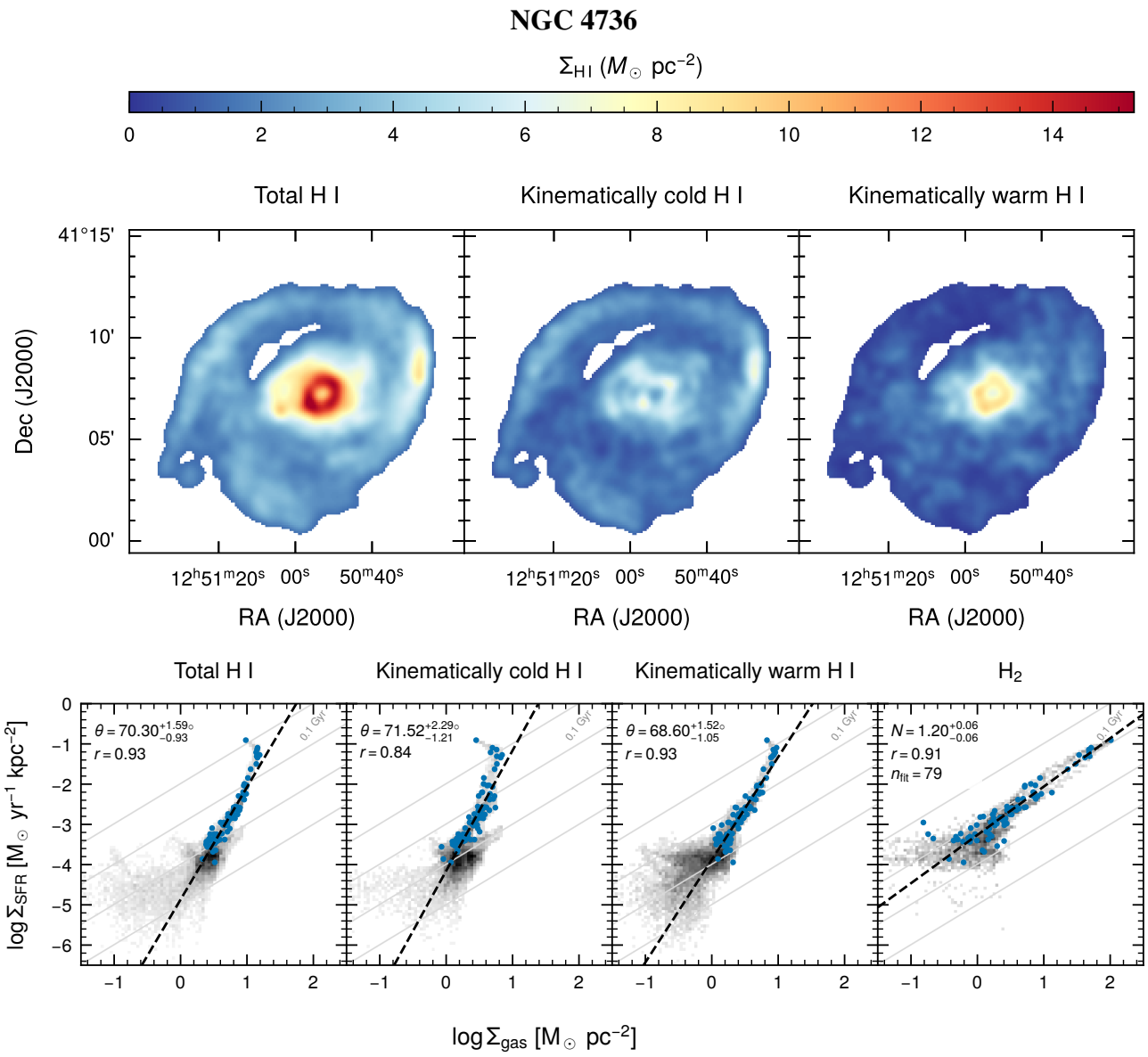


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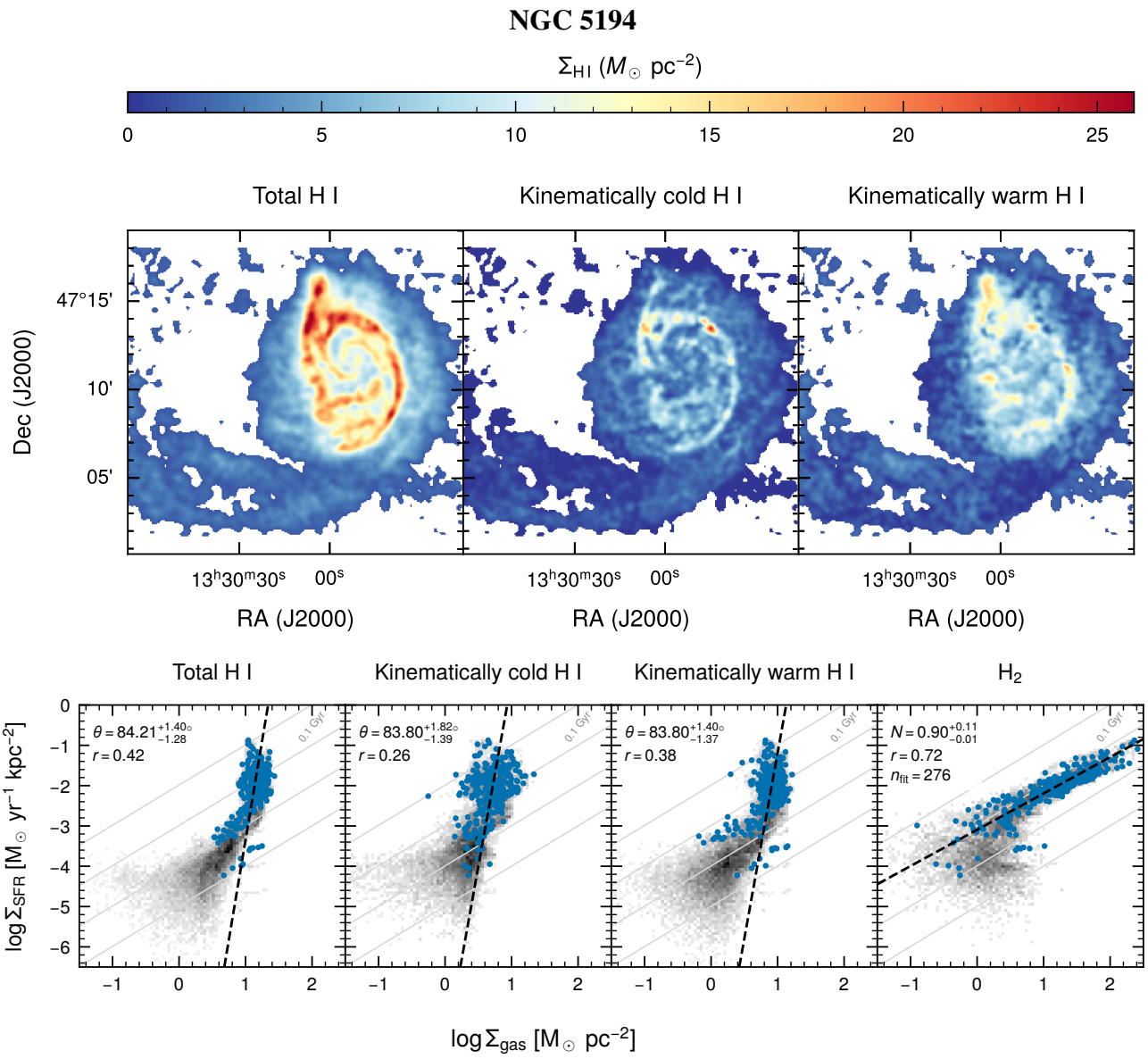


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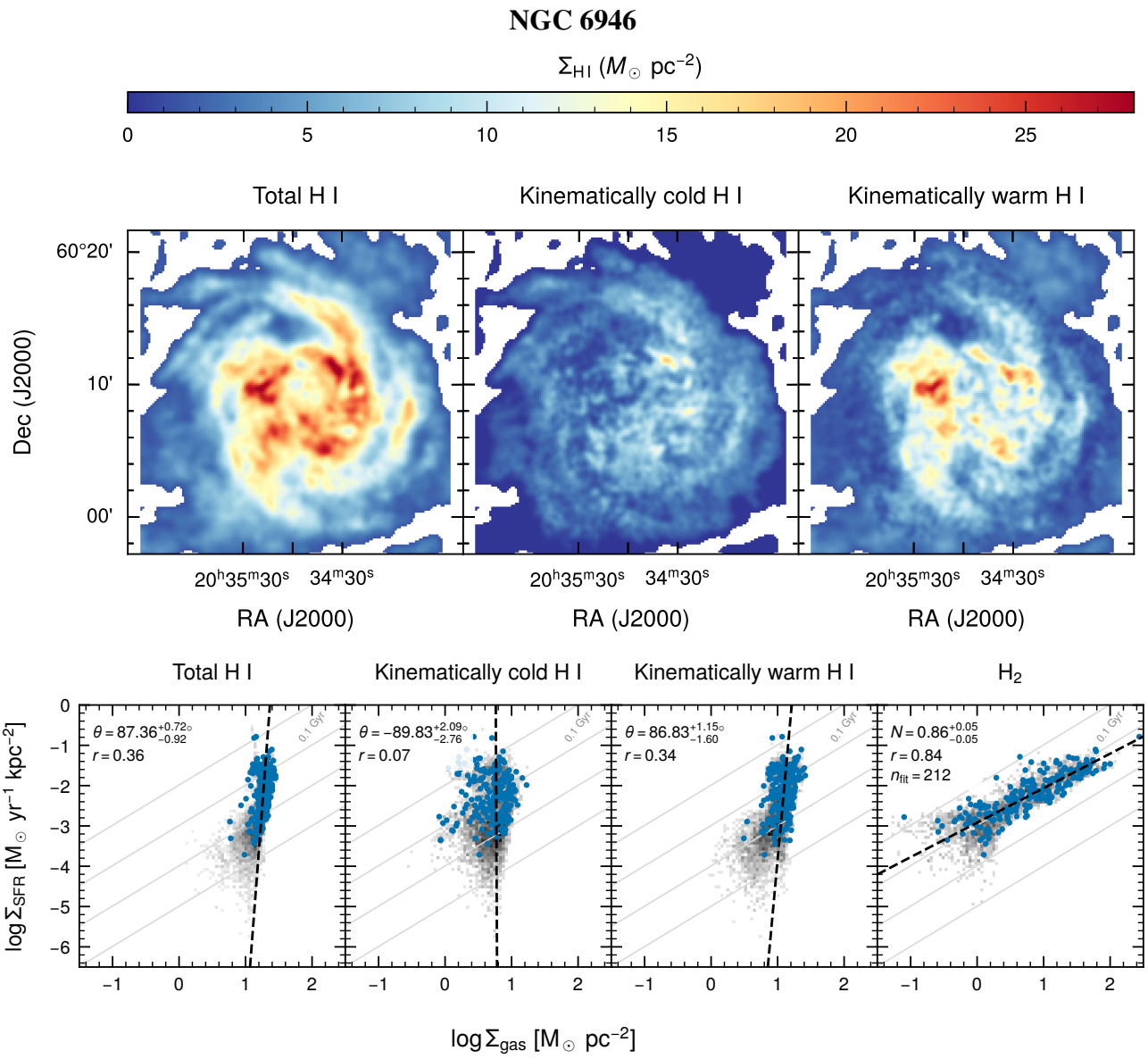


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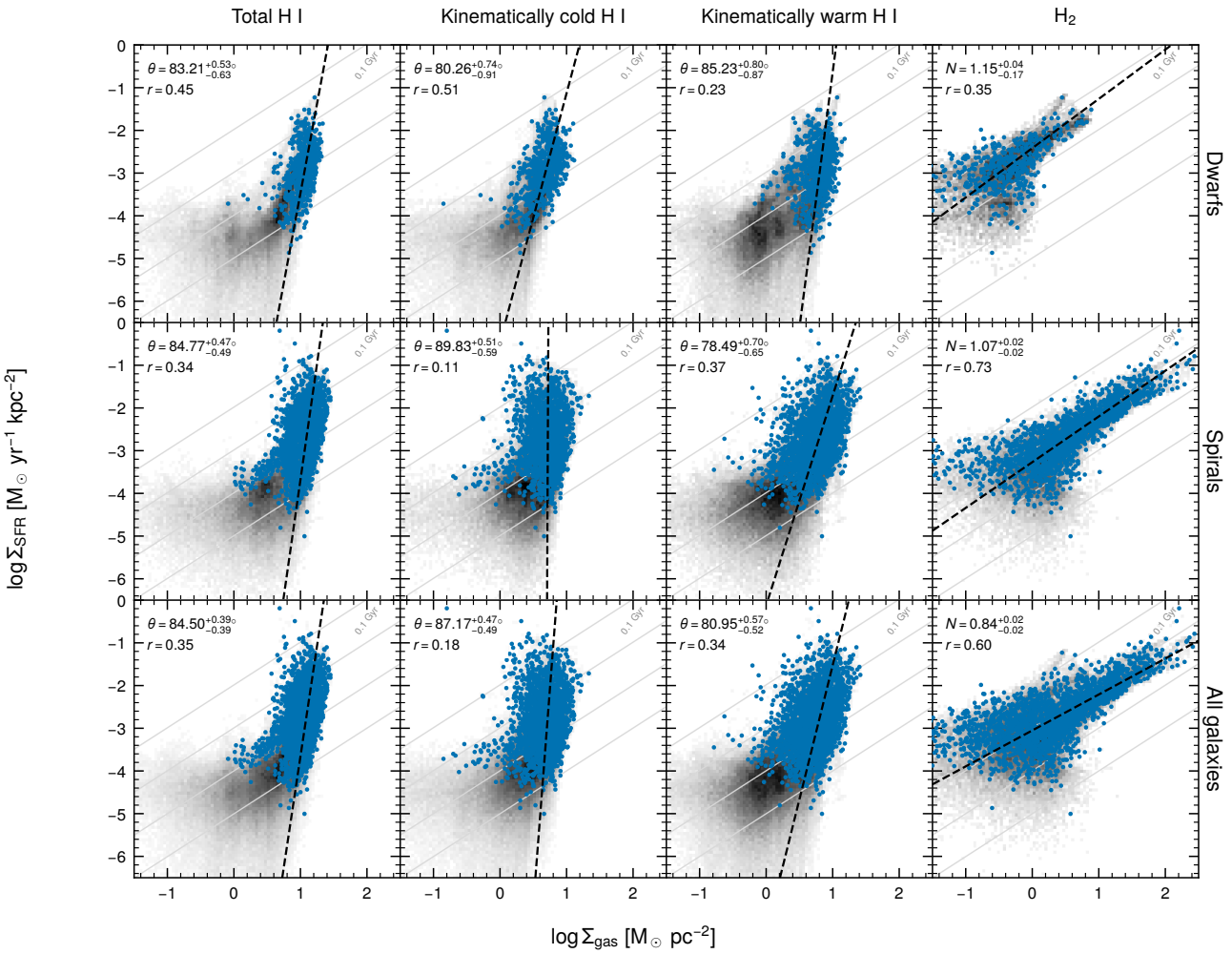


Figure 6: Combined Kennicutt–Schmidt relations for dwarfs (top), spirals (middle) and all galaxies (bottom), at a common physical resolution of 800 pc. The columns show total, kinematically cold and kinematically warm HI, and H_2, from left to right. Symbols and density-weighted fits follow Figure 5. Fits to the HI–SFR relations show θ, and fits to the H_2–SFR relation show N.

3.3 Star formation efficiency

The Kennicutt–Schmidt relations show how the gas and SFR surface densities vary together. Following Leroy et al. (2008), we ask how the apparent SFE differs between the narrow and broad HI reservoirs, and how it changes across the disc.

We calculate the SFE from the new narrow and broad maps, using the same matched spatial resolution within each galaxy as in the individual Kennicutt–Schmidt comparisons. For a region containing N_{reg} valid pixels, we use

$$\mathrm{SFE}_p = 10^{-6} \frac{\sum_{i=1}^{N_{\mathrm{reg}}} \Sigma_{\mathrm{SFR},i} A_i}{\sum_{i=1}^{N_{\mathrm{reg}}} \Sigma_{p,i} A_i}, \tag{3.1}$$

where A_i is the pixel area and p denotes the gas tracer. The factor 10^{-6} converts the SFR surface density from per kpc^2 to per pc^2, giving SFE in yr^{-1}. To compare the tracers over the same area, we calculate their SFEs from the same pixel values in the gas and SFR maps.

Figure 7 shows the SFE as a function of radius for the different gas components. Each point represents the median SFE of the galaxies contributing to a radial bin of width $0.1 r_{25}$. In dwarfs, the narrow HI reservoir appears more efficient at forming stars than the broad HI reservoir throughout the disc. In spirals, narrow HI has the higher SFE in the inner disc ($r < 0.5 r_{25}$), while the efficiencies of the narrow and broad components become more similar farther out. In both samples, the apparent

Table 4: ***Median aperture*** $\log_{10}$**SFE,** *in* yr^{-1}*, calculated separately for each galaxy before combining the sample. Intervals are the central 68% of 10 000 whole-galaxy bootstrap resamples; calibration and decomposition uncertainties are not included. Atomic rows use nine dwarfs and seven spirals. Molecular rows use the same CO-covered pixels within four dwarfs and seven spirals.*

Gas denominator	Dwarfs	Spirals	All galaxies
Total HI	$-9.69^{+0.21}_{-0.10}$	$-9.16^{+0.17}_{-0.13}$	$-9.39^{+0.14}_{-0.29}$
Narrow HI	$-9.29^{+0.21}_{-0.15}$	$-8.66^{+0.05}_{-0.33}$	$-9.08^{+0.19}_{-0.28}$
Broad HI	$-9.47^{+0.20}_{-0.06}$	$-8.99^{+0.24}_{-0.00}$	$-9.13^{+0.14}_{-0.27}$
H_2	$-8.45^{+0.19}_{-0.09}$	$-9.10^{+0.13}_{-0.13}$	$-8.96^{+0.39}_{-0.15}$
H_2 + narrow HI	$-9.02^{+0.13}_{-0.23}$	$-9.25^{+0.02}_{-0.10}$	$-9.23^{+0.17}_{-0.12}$
H_2 + broad HI	$-8.99^{+0.04}_{-0.27}$	$-9.35^{+0.09}_{-0.03}$	$-9.26^{+0.25}_{-0.13}$
H_2 + total HI	$-9.22^{+0.06}_{-0.28}$	$-9.43^{+0.01}_{-0.02}$	$-9.42^{+0.20}_{-0.03}$

SFE decreases towards the outer disc for total HI and for either atomic component, consistent with the radial decline found in HI-dominated regions by Leroy et al. (2008).

The molecular SFE is comparatively flat with radius in spirals, while adding either atomic component introduces a radial decline. Adding HI also lowers the SFE values. The H_2+narrow and H_2+broad profiles are similar.

In the innermost regions of spirals, the H_2+narrow and H_2+broad SFEs are close to the molecular-only SFE because the gas in these CO-covered regions is predominantly molecular. Adding either atomic component therefore makes little difference. Farther out, the atomic contribution increases, so adding HI lowers the SFE more and the combined profiles diverge from the molecular-only profile, consistent with the radial decrease in molecular fraction found by Leroy et al. (2008). In the CO-covered dwarfs, the H_2+narrow and H_2+broad SFEs are lower than the molecular-only SFE even in the central regions. Unlike spiral centres, these regions already contain a substantial atomic-gas contribution. As this contribution increases outwards, the two combined profiles fall farther below the molecular-only profile.

Table 4 summarises the median SFEs of the dwarf and spiral samples.

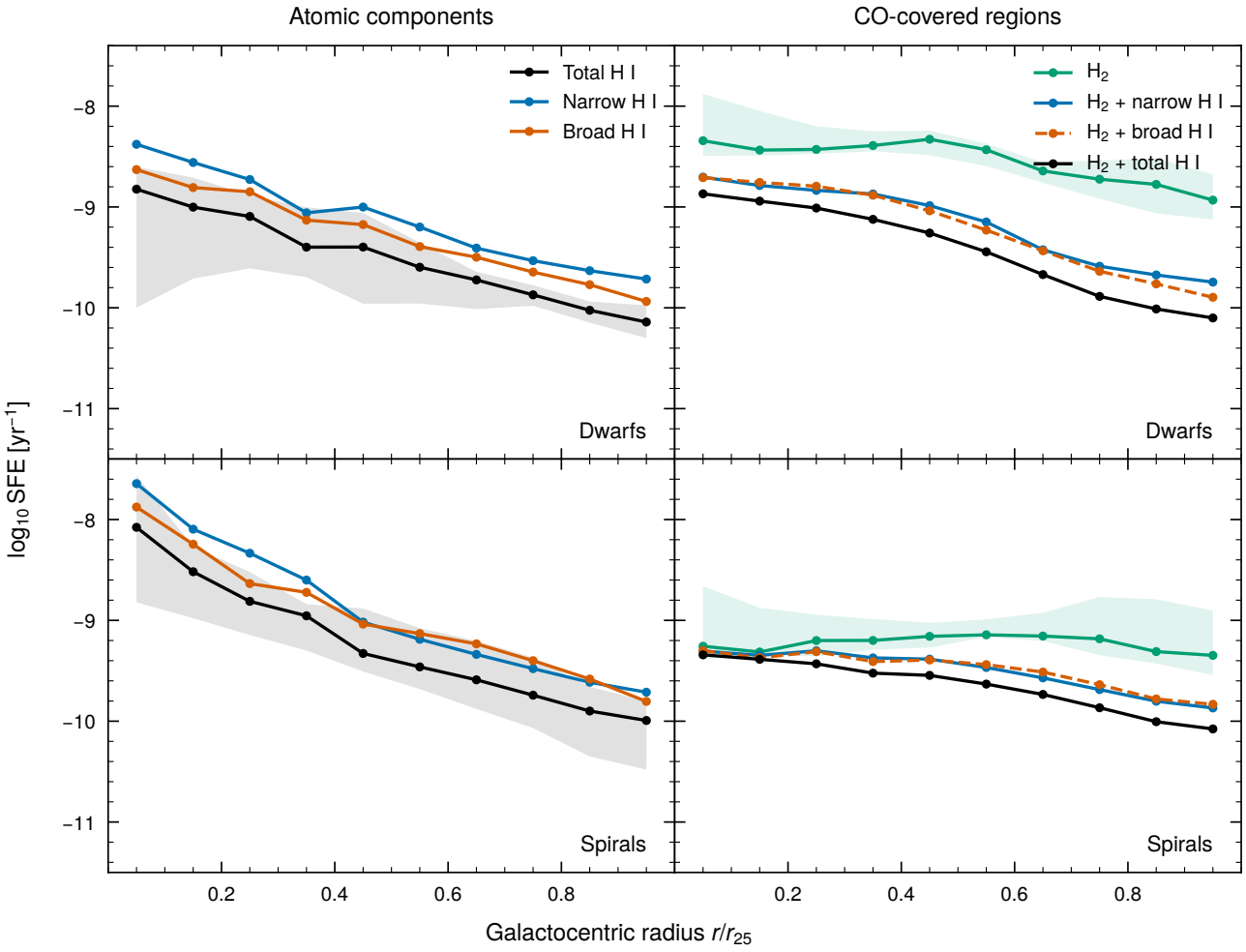


Figure 7: Radial SFE profiles for dwarfs (top) and spirals (bottom). Left: the observed SFR divided by total, narrow or broad H I*. Right:* H_2 *and its combinations with either atomic component or total* H I*, over identical CO-covered positions. Each point is the median of the galaxy measurements in a* $0.1r_{25}$ *bin. Shading shows the 16th–84th percentile spread between galaxies for total* H I *(left) and* H_2 *(right), rather than uncertainty on the median.*

Chapter 4: Conclusion

This thesis examined how kinematically narrow and broad HI are distributed and associated with star formation in 16 nearby galaxies. We used ROHSA to decompose the HI profiles and compared the separated maps with CO-based H_2 and SFR maps. We use the kinematically cold and warm HI components as proxies for the CNM and WNM.

Both atomic components extend across much of the mapped HI discs. Across the sample, the broad component contributes a median of about 55% of the total HI mass, while the kinematically cold component accounts for the remaining 45%. The kinematically cold component has a more filamentary appearance, but its fractional contribution is not consistently higher in stellar spiral arms. We also find no clear evidence that narrow HI is generally more concentrated near CO emission than broad HI. This may be explained by how atomic gas behaves during the atomic-to-molecular transition where atomic gas remains in shielding layers around molecular interiors while additional gas becomes molecular. Hydrogen also no longer contributes to the HI emission once it is converted into H_2. Additionally, H_2 can survive where CO is dissociated, so our CO-faint regions are not necessarily free of H_2.

We examined the Kennicutt–Schmidt relations for the different gas components. Their behaviour depends on the galaxy population. In the combined dwarf sample, narrow HI correlates more strongly with the SFR than total or broad HI. In spirals, broad HI follows the SFR more closely than narrow HI. One possible explanation for the stronger broad-component relation in spirals is that turbulence broadens the emission of cold gas near star-forming regions, allowing some of it to be assigned to the broad component.

As already established in the literature, the molecular component remains more closely associated with the SFR in most spirals and follows an approximately linear Kennicutt–Schmidt relation (Bigiel et al., 2008).

The SFE of both the narrow and broad HI components declines towards the outer disc. In dwarfs, the narrow HI component has a higher SFE than the broad component across the disc. In spirals, narrow HI has the higher SFE in the inner disc, while the efficiencies of the narrow and broad components become more similar farther out. One possible explanation for this convergence is that turbulent broadening becomes weaker towards the outer disc, consistent with the radial decline in HI line widths reported by Tamburro et al. (2009). A larger fraction of the cold gas may then contribute to the narrow

component rather than the broad component, bringing their relative contributions and measured SFEs closer together.

Our data do not allow us to locate where the HI-to-H_2 transition occurs, since molecular hydrogen can be present without detectable CO emission. Extending the analysis to a larger sample and incorporating data from more recent surveys with improved spatial and spectral resolution, independent constraints on the thermal phases and molecular-gas measurements, accounting for metallicity or other parameters would help clarify the role of cold atomic gas. A larger dwarf sample would also help establish whether the stronger combined narrow-HI relation is a general property of these galaxies.

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Appendix A: ROHSA parameters

Table 5 gives the settings used for the final ROHSA decompositions described in Section 2.4. The amplitude and global dispersion penalties are zero, while the centroid and local dispersion penalties are both one. Each galaxy is fitted with two or three Gaussian components per spectrum. These Gaussians describe the shape of the emission profile; their number does not represent the number of thermal phases. The minimum fitted dispersion is $\sigma_{\min} = 1.274\,\Delta v$, where Δv is the velocity-channel spacing. This corresponds to three channels across the Gaussian FWHM and prevents the fit from assigning widths narrower than this adopted sampling limit. The bound therefore differs between galaxies because their channel spacings differ.

Table 5: ***Settings common to the adopted ROHSA fits****. The code names identify the corresponding input parameters. Dispersion quantities are shown in physical units; ROHSA receives these values in channels.*

Setting	Code parameter	Adopted value
Amplitude regularisation	`lambda_amp`	0
Centroid regularisation	`lambda_mu`	1
Dispersion regularisation	`lambda_sig`	1
Global width penalty	`lambda_var_sig`	0
Other variance penalties	`lambda_var_amp`, `lambda_var_mu`	0
Additional width penalty	`lambda_lym_sig`	0 (disabled)
Minimum fitted dispersion	`lb_sig`	$1.274\,\Delta v$
Maximum fitted dispersion	`ub_sig`	40 km s^{-1}
Initial dispersion	`sig_init`	$\max(8\,\mathrm{km\,s^{-1}}, 1.5\sigma_{\min})$
Initial dispersion bounds	`lb_sig_init`, `ub_sig_init`	$\sigma_{\min}$, 20 km s^{-1}
Initialisation spectrum	`init_option`	Mean spectrum
Initial amplitude factor	`amp_fact_init`	0.66
Initialisation optimiser limit	`maxiter_init`	15 000
Fitting optimiser limit	`maxiter`	800
Optimiser memory	`m`	10
Supplied noise map	`noise`	Enabled
Spatial regularisation	`regul`	Enabled
Multiresolution fitting	`descent`	Enabled

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La répartition de l'hydrogène atomique (HI) cinématiquement froid et chaud et son lien avec la formation stellaire dans les galaxies proches

Résumé

Le processus de formation stellaire au sein des galaxies est alimenté par le gaz du milieu interstellaire (MIS), qui comprend des phases moléculaire, atomique et ionisée. Cette étude examine la relation entre la formation stellaire et les différentes phases du MIS dans un échantillon de 16 galaxies proches, dont 9 naines/de type tardif et 7 spirales. Nous avons utilisé les données CO de *HERACLES*, les données d'hydrogène atomique (HI) de *THINGS*, les données à 24 μm de *SINGS*, ainsi que les données UV de *GALEX NGS*. Nous avons décomposé les spectres d'émission de l'HI à l'aide de l'algorithme ROHSA, qui prend en compte la cohérence spatiale des spectres voisins, afin de séparer les composantes HI cinématiquement étroites et larges. Nous utilisons ces composantes comme traceurs cinématiques du milieu neutre froid (CNM) et du milieu neutre tiède (WNM), plutôt que comme des mesures directes des phases thermiques.

Nous examinons la répartition spatiale des composantes cinématiquement froide et chaude et constatons qu'elles s'étendent toutes deux jusqu'aux régions les plus externes des disques atomiques observés. La composante étroite représente en médiane environ 45% de la masse totale d'HI et présente un aspect plus filamentaire que la composante large. Sa fraction n'est pas systématiquement plus élevée dans les bras spiraux stellaires, contrairement aux attentes théoriques selon lesquelles le refroidissement dans les bras pourrait favoriser la formation de gaz froid. De plus, nous ne trouvons pas d'indication claire que la composante étroite de l'HI soit généralement plus concentrée au voisinage de l'émission CO que la composante large, comme on pourrait s'y attendre si la composante étroite était plus étroitement associée au gaz moléculaire.

Nous examinons la relation de Kennicutt–Schmidt pour les différentes composantes gazeuses afin de déterminer lesquelles sont les plus étroitement liées à la formation stellaire. Dans les galaxies naines, la composante cinématiquement froide présente une corrélation globale plus forte avec le taux de formation stellaire que l'HI total ou la composante chaude. Dans les galaxies spirales, la composante chaude est globalement mieux corrélée au taux de formation stellaire que la composante froide. Cette différence pourrait s'expliquer par le fait que la largeur des raies renseigne sur des conditions du gaz différentes dans ces deux types de galaxies. Dans les naines, l'émission étroite trace probablement du gaz plus froid et moins turbulent, associé à des conditions favorables à la formation stellaire. Dans les spirales, les régions de formation stellaire peuvent contenir du gaz froid dont les raies sont élargies par la turbulence ou des mouvements non résolus. Globalement, le gaz moléculaire reste la composante la plus étroitement liée au taux de formation stellaire, comme l'ont montré les études antérieures. Le rôle du gaz atomique cinématiquement froid dans la transition HI–H_2 reste ambigu à notre résolution sub-kiloparsec.

Mots-clés : Galaxies proches, Milieu interstellaire, Formation stellaire

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The Distribution of Kinematically Cold and Warm Atomic Hydrogen (HI) and Its Connection to Star Formation in Nearby Galaxies

Abstract

The process of star formation within galaxies is fueled by gas from the interstellar medium (ISM), which exists in molecular, atomic and ionised phases. This study investigates the relationship between star formation and the different phases of the ISM in a sample of 16 nearby galaxies, composed of 9 late-type/dwarfs and 7 spirals. We used CO data from *HERACLES*, atomic hydrogen (HI) data from *THINGS*, 24 μm data from *SINGS*, and UV data from *GALEX NGS* surveys. We decomposed the HI emission spectra using the ROHSA algorithm, which takes into account the spatial coherence of neighbouring spectra, to separate kinematically narrow and broad HI components. We use these components as kinematic proxies for the cold neutral medium (CNM) and warm neutral medium (WNM), rather than direct measurements of the thermal phases.

We examine the spatial distribution of the kinematically cold and warm components and find that both components extend across the outermost observed atomic discs. The narrow component contains a median of about 45% of the total HI mass and has a more filamentary appearance than the broad component. Its fractional contribution is not consistently higher in stellar spiral arms, contrary to theoretical expectations that cooling in the arms could favour cold gas formation. In addition, we find no clear evidence that narrow HI is generally more concentrated near CO emission than broad HI, as would be expected if the narrow component were more closely associated with molecular gas.

We examine the Kennicutt–Schmidt relation for the different gas components to investigate which components are more closely linked to star formation. In dwarf galaxies, the kinematically cold component shows a stronger overall correlation with the SFR than the total HI or the warm component. In spiral galaxies, the warm component is more globally correlated with the SFR than the cold component. A possible explanation of this discrepancy is that line width traces different gas conditions in the two types of galaxies. In dwarfs, narrow emission likely traces cooler, less turbulent gas associated with conditions favourable for star formation. In spirals, star-forming regions can contain cold gas whose lines are broadened by turbulence or unresolved motions. Overall, molecular gas remains the component that is more closely linked to the SFR, as previously shown in the literature. The role of kinematically cold atomic gas in the HI-to-H_2 transition remains ambiguous at our sub-kpc resolution.

Keywords: Nearby galaxies, Interstellar medium, Star formation